

Model for “Th1”

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General Condition

- Basis type: **lgs**
- SAMB selection:
 - Type: [Q, G]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u , T_g , T_u]
 - Spin (s): [0, 1]
- Atomic selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u , T_g , T_u]
 - Spin (s): [0, 1]
- Site-cluster selection:
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u , T_g , T_u]
- Bond-cluster selection:
 - Type: [Q, G, M, T]
 - Rank: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]
 - Irrep.: [A_g , A_u , E_g , E_u , T_g , T_u]
- Max. neighbor: 10
- Search cell range: (-2, 3), (-2, 3), (-2, 3)
- Toroidal priority: **false**

Group and Unit Cell

- Group: SG No. 200 $T_h^1 Pm\bar{3}$ [cubic]
- Associated point group: PG No. 200 $T_h m\bar{3}$ [cubic]
- Unit cell:
 - $a = 1.00000$, $b = 1.00000$, $c = 1.00000$, $\alpha = 90.0$, $\beta = 90.0$, $\gamma = 90.0$
- Lattice vectors (conventional cell):
 - $\mathbf{a}_1 = [1.00000, 0.00000, 0.00000]$
 - $\mathbf{a}_2 = [0.00000, 1.00000, 0.00000]$
 - $\mathbf{a}_3 = [0.00000, 0.00000, 1.00000]$

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1 0\}$	2	$\{2_{001} 0\}$	3	$\{2_{010} 0\}$	4	$\{2_{100} 0\}$	5	$\{3_{111}^+ 0\}$
6	$\{3_{-11-1}^+ 0\}$	7	$\{3_{1-1-1}^+ 0\}$	8	$\{3_{-1-11}^+ 0\}$	9	$\{3_{111}^- 0\}$	10	$\{3_{1-1-1}^- 0\}$
11	$\{3_{-1-11}^- 0\}$	12	$\{3_{-11-1}^- 0\}$	13	$\{-1 0\}$	14	$\{m_{001} 0\}$	15	$\{m_{010} 0\}$
16	$\{m_{100} 0\}$	17	$\{-3_{111}^+ 0\}$	18	$\{-3_{-11-1}^+ 0\}$	19	$\{-3_{1-1-1}^+ 0\}$	20	$\{-3_{-1-11}^+ 0\}$
21	$\{-3_{111}^- 0\}$	22	$\{-3_{1-1-1}^- 0\}$	23	$\{-3_{-1-11}^- 0\}$	24	$\{-3_{-11-1}^- 0\}$		

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_g)$	A_g	0	Q, T	-	-	1
2	$\mathbb{G}_3(A_g)$	A_g	3	G, M	-	-	$\sqrt{15}xyz$
3	$\mathbb{Q}_4(A_g)$	A_g	4	Q, T	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$
4	$\mathbb{G}_0(A_u)$	A_u	0	G, M	-	-	1
5	$\mathbb{Q}_3(A_u)$	A_u	3	Q, T	-	-	$\sqrt{15}xyz$
6	$\mathbb{G}_4(A_u)$	A_u	4	G, M	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
7	$\mathbb{Q}_{2,1}(E_g)$	E_g	2	Q, T	-	1	$\frac{-x^2-y^2+2z^2}{2}$
8	$\mathbb{Q}_{2,2}(E_g)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
9	$\mathbb{Q}_{4,1}(E_g)$	E_g	4	Q, T	-	1	$\frac{\sqrt{15}(-x^4+12x^2y^2-6x^2z^2-y^4-6y^2z^2+2z^4)}{12}$
10	$\mathbb{Q}_{4,2}(E_g)$					2	$\frac{\sqrt{5}(x^4-6x^2z^2-y^4+6y^2z^2)}{4}$
11	$\mathbb{G}_{2,1}(E_u)$	E_u	2	G, M	-	1	$\frac{-x^2-y^2+2z^2}{2}$
12	$\mathbb{G}_{2,2}(E_u)$					2	$\frac{\sqrt{3}(x^2-y^2)}{2}$
13	$\mathbb{G}_{4,1}(E_u)$	E_u	4	G, M	-	1	$\frac{\sqrt{15}(-x^4+12x^2y^2-6x^2z^2-y^4-6y^2z^2+2z^4)}{12}$
14	$\mathbb{G}_{4,2}(E_u)$					2	$\frac{\sqrt{5}(x^4-6x^2z^2-y^4+6y^2z^2)}{4}$
15	$\mathbb{Q}_{5,1}(E_u)$	E_u	5	Q, T	-	1	$\frac{3\sqrt{35}xyz(x^2-y^2)}{2}$
16	$\mathbb{Q}_{5,2}(E_u)$					2	$\frac{\sqrt{105}xyz(x^2+y^2-2z^2)}{2}$
17	$\mathbb{G}_{1,1}(T_g)$	T_g	1	G, M	-	1	x
18	$\mathbb{G}_{1,2}(T_g)$					2	y
19	$\mathbb{G}_{1,3}(T_g)$					3	z
20	$\mathbb{Q}_{2,1}(T_g)$	T_g	2	Q, T	-	1	$\sqrt{3}yz$
21	$\mathbb{Q}_{2,2}(T_g)$					2	$\sqrt{3}xz$
22	$\mathbb{Q}_{2,3}(T_g)$					3	$\sqrt{3}xy$
23	$\mathbb{G}_{3,1}(T_g, 1)$	T_g	3	G, M	1	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
24	$\mathbb{G}_{3,2}(T_g, 1)$					2	$\frac{y(-3x^2+2y^2-3z^2)}{2}$
25	$\mathbb{G}_{3,3}(T_g, 1)$					3	$\frac{z(-3x^2-3y^2+2z^2)}{2}$
26	$\mathbb{G}_{3,1}(T_g, 2)$	T_g	3	G, M	2	1	$\frac{\sqrt{15}x(y^2-z^2)}{2}$
27	$\mathbb{G}_{3,2}(T_g, 2)$					2	$\frac{\sqrt{15}y(-x^2+z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
28	$\mathbb{G}_{3,3}(T_g, 2)$					3	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
29	$\mathbb{Q}_{4,1}(T_g, 1)$	T_g	4	Q, T	1	1	$\frac{\sqrt{35}yz(y^2-z^2)}{2}$
30	$\mathbb{Q}_{4,2}(T_g, 1)$					2	$\frac{\sqrt{35}xz(-x^2+z^2)}{2}$
31	$\mathbb{Q}_{4,3}(T_g, 1)$					3	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
32	$\mathbb{Q}_{4,1}(T_g, 2)$	T_g	4	Q, T	2	1	$\frac{\sqrt{5}yz(6x^2-y^2-z^2)}{2}$
33	$\mathbb{Q}_{4,2}(T_g, 2)$					2	$\frac{\sqrt{5}xz(-x^2+6y^2-z^2)}{2}$
34	$\mathbb{Q}_{4,3}(T_g, 2)$					3	$\frac{\sqrt{5}xy(-x^2-y^2+6z^2)}{2}$
35	$\mathbb{Q}_{1,1}(T_u)$	T_u	1	Q, T	-	1	x
36	$\mathbb{Q}_{1,2}(T_u)$					2	y
37	$\mathbb{Q}_{1,3}(T_u)$					3	z
38	$\mathbb{G}_{2,1}(T_u)$	T_u	2	G, M	-	1	$\sqrt{3}yz$
39	$\mathbb{G}_{2,2}(T_u)$					2	$\sqrt{3}xz$
40	$\mathbb{G}_{2,3}(T_u)$					3	$\sqrt{3}xy$
41	$\mathbb{Q}_{3,1}(T_u, 1)$	T_u	3	Q, T	1	1	$\frac{x(2x^2-3y^2-3z^2)}{2}$
42	$\mathbb{Q}_{3,2}(T_u, 1)$					2	$\frac{y(-3x^2+2y^2-3z^2)}{2}$
43	$\mathbb{Q}_{3,3}(T_u, 1)$					3	$\frac{z(-3x^2-3y^2+2z^2)}{2}$
44	$\mathbb{Q}_{3,1}(T_u, 2)$	T_u	3	Q, T	2	1	$\frac{\sqrt{15}x(y^2-z^2)}{2}$
45	$\mathbb{Q}_{3,2}(T_u, 2)$					2	$\frac{\sqrt{15}y(-x^2+z^2)}{2}$
46	$\mathbb{Q}_{3,3}(T_u, 2)$					3	$\frac{\sqrt{15}z(x^2-y^2)}{2}$
47	$\mathbb{G}_{4,1}(T_u, 1)$	T_u	4	G, M	1	1	$\frac{\sqrt{35}yz(y^2-z^2)}{2}$
48	$\mathbb{G}_{4,2}(T_u, 1)$					2	$\frac{\sqrt{35}xz(-x^2+z^2)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
49	$\mathbb{G}_{4,3}(T_u, 1)$					3	$\frac{\sqrt{35}xy(x^2-y^2)}{2}$
50	$\mathbb{G}_{4,1}(T_u, 2)$	T_u	4	G, M	2	1	$\frac{\sqrt{5}yz(6x^2-y^2-z^2)}{2}$
51	$\mathbb{G}_{4,2}(T_u, 2)$					2	$\frac{\sqrt{5}xz(-x^2+6y^2-z^2)}{2}$
52	$\mathbb{G}_{4,3}(T_u, 2)$					3	$\frac{\sqrt{5}xy(-x^2-y^2+6z^2)}{2}$
53	$\mathbb{Q}_{5,1}(T_u, 3)$	T_u	5	Q, T	3	1	$\frac{\sqrt{105}x(2x^2y^2-2x^2z^2-y^4+z^4)}{4}$
54	$\mathbb{Q}_{5,2}(T_u, 3)$					2	$\frac{\sqrt{105}y(x^4-2x^2y^2+2y^2z^2-z^4)}{4}$
55	$\mathbb{Q}_{5,3}(T_u, 3)$					3	$\frac{\sqrt{105}z(-x^4+2x^2z^2+y^4-2y^2z^2)}{4}$

Basis in full matrix

Table 3: dimension = 24

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$ s, \uparrow\rangle @A(1)$	1	$ s, \downarrow\rangle @A(1)$	2	$ p_x, \uparrow\rangle @A(1)$	3	$ p_x, \downarrow\rangle @A(1)$	4	$ p_y, \uparrow\rangle @A(1)$
5	$ p_y, \downarrow\rangle @A(1)$	6	$ p_z, \uparrow\rangle @A(1)$	7	$ p_z, \downarrow\rangle @A(1)$	8	$ s, \uparrow\rangle @A(2)$	9	$ s, \downarrow\rangle @A(2)$
10	$ p_x, \uparrow\rangle @A(2)$	11	$ p_x, \downarrow\rangle @A(2)$	12	$ p_y, \uparrow\rangle @A(2)$	13	$ p_y, \downarrow\rangle @A(2)$	14	$ p_z, \uparrow\rangle @A(2)$
15	$ p_z, \downarrow\rangle @A(2)$	16	$ s, \uparrow\rangle @A(3)$	17	$ s, \downarrow\rangle @A(3)$	18	$ p_x, \uparrow\rangle @A(3)$	19	$ p_x, \downarrow\rangle @A(3)$
20	$ p_y, \uparrow\rangle @A(3)$	21	$ p_y, \downarrow\rangle @A(3)$	22	$ p_z, \uparrow\rangle @A(3)$	23	$ p_z, \downarrow\rangle @A(3)$		

Table 4: Atomic basis (orbital part only)

orbital	definition
$ s\rangle$	1
$ p_x\rangle$	x
$ p_y\rangle$	y
$ p_z\rangle$	z

SAMB: 1428 (all 1428)

• **A : 'A' site-cluster**

- * bra: $\langle s, \uparrow |, \langle s, \downarrow |$
- * ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$
- * wyckoff: **3d**

$$\boxed{\text{z1}} \quad \mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z135}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z136}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

• **A : 'A' site-cluster**

- * bra: $\langle s, \uparrow |, \langle s, \downarrow |$
- * ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$
- * wyckoff: **3d**

$$\boxed{\text{z70}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u) = \frac{\sqrt{2}\left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g)\right)}{2}$$

$$\boxed{\text{z71}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(s)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(s)}(E_g) \right)}{2}$$

$$\boxed{\text{z72}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z273}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z274}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z275}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(s)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z276}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(s)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z277}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2} \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z278}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2} \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z880}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z881}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z882}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z883}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, b) = \frac{-\sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(s)}(E_g) + 3 \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z884}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, b) = -\frac{\sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(s)}(E_g) + 3 \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z885}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{3} \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$$

$$\begin{aligned}
\text{z886} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2) &= -\frac{3\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) + \sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z887} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2) &= \frac{3\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z888} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3} \\
\text{z889} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) &= \frac{3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z890} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) &= \frac{-3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z891} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) &= -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3} \\
\text{z892} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) &= \frac{-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z893} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) &= -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z894} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3} \\
\text{z895} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3} \\
\text{z896} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3} \\
\text{z897} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3} \\
\text{z898} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, b) &= \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) + 3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z899} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, b) &= -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}
\end{aligned}$$

$$\boxed{\text{z900}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$$

$$\boxed{\text{z901}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) = -\frac{3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) + \sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z902}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) = \frac{3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z903}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3}$$

$$\boxed{\text{z904}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z905}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z906}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

• **A : 'A' site-cluster**

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: **3d**

$$\boxed{\text{z2}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z3}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g) \right)}{2}$$

$$\boxed{\text{z4}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g) \right)}{2}$$

$$\boxed{\text{z5}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g) = \mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_0^{(s)}(A_g)$$

$$\boxed{\text{z6}} \quad \mathbb{G}_3^{(c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g) \right)}{2}$$

$$\boxed{\text{z7}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(s)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(s)}(E_g) \right)}{2}$$

$$\boxed{\text{z137}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z138}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z139}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z140}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z141}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(s)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z142}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(s)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z143}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z144}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(s)}(A_g)}{2}$$

$$\boxed{\text{z145}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(s)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z146}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(s)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z147}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z148}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$$\boxed{\text{z403}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z404}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z405}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z406}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z407}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z408}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$$

$$\boxed{\text{z409}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 1) = -\frac{3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z410}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z411}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3}$$

$$\boxed{\text{z412}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z413}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z414}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3}$$

$$\boxed{\text{z415}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\boxed{\text{z416}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$$\begin{aligned}
\text{z417} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) &= -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3} \\
\text{z418} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1) &= -\frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z419} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1) &= \frac{3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z420} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3} \\
\text{z421} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g) &= -\frac{3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + \sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z422} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g) &= \frac{3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z423} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3} \\
\text{z424} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3} \\
\text{z425} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3} \\
\text{z426} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(s)}(A_g)}{3} \\
\text{z427} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, b) &= \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z428} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, b) &= -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6} \\
\text{z429} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}
\end{aligned}$$

• A;A_001_1 : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$
 * ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$
 * wyckoff: 12a@12j

$$\boxed{\text{z8}} \quad \mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z9}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z73}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u, a) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3}$$

$$\boxed{\text{z74}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u, b) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3}$$

$$\boxed{\text{z149}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z150}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z151}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z152}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z279}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z280}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$$

$$\boxed{\text{z281}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z282}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$$

$$\boxed{\text{z430}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z431}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z432}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z433}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = -\frac{3\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z434}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{3\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z435}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z436}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z437}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z438}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z439}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z440}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z441}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z442}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, b) = \frac{-\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z443}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z444}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z445}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, c) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)\right)}{6}$$

$$\boxed{\text{z446}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, c) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6}$$

$$\boxed{\text{z447}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, c) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6}$$

$$\boxed{\text{z907}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z908}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z909}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z910}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z911}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z912}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z913}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)\right)}{6}$$

$$\boxed{\text{z914}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)\right)}{6}$$

$$\text{z915} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\text{z916} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6}$$

$$\text{z917} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\text{z918} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\text{z919} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6}$$

$$\text{z920} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\text{z921} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\text{z922} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6}$$

$$\text{z923} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\text{z924} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

• **A;A_001_1** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

* wyckoff: 12a@12j

$$\text{z10} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\begin{aligned}
\boxed{\text{z11}} \quad \mathbb{Q}_0^{(c)}(A_g, b) &= \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3} \\
\boxed{\text{z12}} \quad \mathbb{Q}_0^{(c)}(A_g, c) &= \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3} \\
\boxed{\text{z13}} \quad \mathbb{Q}_4^{(c)}(A_g) &= \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3} \\
\boxed{\text{z14}} \quad \mathbb{Q}_0^{(1,0;c)}(A_g, a) &= \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3} \\
\boxed{\text{z15}} \quad \mathbb{Q}_0^{(1,0;c)}(A_g, b) &= \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3} \\
\boxed{\text{z16}} \quad \mathbb{Q}_0^{(1,0;c)}(A_g, c) &= \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3} \\
\boxed{\text{z17}} \quad \mathbb{Q}_4^{(1,0;c)}(A_g) &= \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3} \\
\boxed{\text{z18}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, a) &= \frac{\sqrt{3} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3} \\
\boxed{\text{z19}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, b) &= - \frac{\sqrt{3} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3} \\
\boxed{\text{z20}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, c) &= \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3} \\
\boxed{\text{z21}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, d) &= \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3} \\
\boxed{\text{z75}} \quad \mathbb{Q}_3^{(c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{3} \\
\boxed{\text{z76}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, a) &= \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}
\end{aligned}$$

$$\boxed{\text{z77}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, b) = \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{2,2}^{(b)}(E_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z78}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, c) = \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z79}} \quad \mathbb{Q}_3^{(1,0;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z80}} \quad \mathbb{G}_0^{(c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z81}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u, a) = \frac{\sqrt{5} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{5}$$

$$\boxed{\text{z82}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u, b) = \frac{\sqrt{2} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{2,1}^{(b)}(E_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z83}} \quad \mathbb{G}_4^{(1,-1;c)}(A_u) = \frac{\sqrt{30} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z84}} \quad \mathbb{G}_0^{(1,0;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z85}} \quad \mathbb{G}_0^{(1,1;c)}(A_u, a) = \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z86}} \quad \mathbb{G}_0^{(1,1;c)}(A_u, b) = \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z153}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{3} \left(-\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z154}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z155}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{3} \left(-\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z156}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$$

$$\boxed{\text{z157}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z158}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$$

$$\boxed{\text{z159}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, d) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z160}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, d) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$$

$$\boxed{\text{z161}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{-\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z162}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{3} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z163}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$$

$$\boxed{\text{z164}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{3} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z165}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, c) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$$

$$\boxed{\text{z166}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, c) = \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z167}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, d) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$$

$$\boxed{\text{z168}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, d) = \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z169}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, a) = \frac{\sqrt{3} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\begin{aligned}
\boxed{\text{z170}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, a) &= \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2} \\
\boxed{\text{z171}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, b) &= \frac{\sqrt{3} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6} \\
\boxed{\text{z172}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, b) &= \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2} \\
\boxed{\text{z173}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, c) &= \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z174}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, c) &= \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{2} \\
\boxed{\text{z175}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, d) &= \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z176}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, d) &= \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{2} \\
\boxed{\text{z283}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, a) &= \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{2} \\
\boxed{\text{z284}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, a) &= \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z285}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, b) &= \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z286}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, b) &= \frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{2} \\
\boxed{\text{z287}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) &= \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z288}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) &= \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2}
\end{aligned}$$

$$\begin{aligned}
\text{z289} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) &= \frac{\sqrt{7} \left(2\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{14} \\
\text{z290} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) &= \frac{\sqrt{7} \left(-2\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{14} \\
\text{z291} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, c) &= \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\text{z292} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, c) &= \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_0^{(b)}(A_g)}{2} \\
\text{z293} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, d) &= \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{2,1}^{(b)}(E_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2} \\
\text{z294} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, d) &= -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{2,2}^{(b)}(E_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} \\
\text{z295} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, e) &= \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{2} \\
\text{z296} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, e) &= \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6} \\
\text{z297} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u) &= \frac{\sqrt{21} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) + 4\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{42} \\
\text{z298} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{14} \\
\text{z299} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, a) &= \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{2} \\
\text{z300} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, a) &= \frac{\sqrt{3} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{6} \\
\text{z301} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, b) &= \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z302}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, b) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z303}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z304}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z305}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, b) = \frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z306}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, b) = \frac{\sqrt{2}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z448}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{6} \left(\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z449}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z450}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z451}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, b) = -\frac{\sqrt{6} \left(\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z452}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, b) = -\frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z453}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, b) = -\frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z454}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z455}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z456}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z457}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z458}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z459}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z460}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 1) = \frac{\sqrt{6} \left(-\mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z461}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z462}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{6} \left(-\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z463}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z464}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(T_u) - \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z465}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{2} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) - 2 \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z466}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) = \frac{-3 \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - 4 \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 4 \sqrt{3} \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{18}$$

$$\boxed{\text{z467}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) = \frac{3 \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 4 \sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 4 \sqrt{3} \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{18}$$

$$\boxed{\text{z468}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{3} \left(-2 \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 2 \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{9}$$

$$\begin{aligned}
\text{z469} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(Eu) \mathbb{T}_{1,1}^{(b)}(Tu, a) + \mathbb{M}_{2,2}^{(1,-1;a)}(Eu) \mathbb{T}_{1,1}^{(b)}(Tu, a) - \mathbb{M}_{2,2}^{(1,-1;a)}(Tu) \mathbb{T}_{1,3}^{(b)}(Tu, a) + \mathbb{M}_{2,3}^{(1,-1;a)}(Tu) \mathbb{T}_{1,2}^{(b)}(Tu, a) \right)}{6} \\
\text{z470} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(Eu) \mathbb{T}_{1,2}^{(b)}(Tu, a) + \mathbb{M}_{2,1}^{(1,-1;a)}(Tu) \mathbb{T}_{1,3}^{(b)}(Tu, a) + \mathbb{M}_{2,2}^{(1,-1;a)}(Eu) \mathbb{T}_{1,2}^{(b)}(Tu, a) - \mathbb{M}_{2,3}^{(1,-1;a)}(Tu) \mathbb{T}_{1,1}^{(b)}(Tu, a) \right)}{6} \\
\text{z471} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(Tu) \mathbb{T}_{1,2}^{(b)}(Tu, a) - 2 \mathbb{M}_{2,2}^{(1,-1;a)}(Eu) \mathbb{T}_{1,3}^{(b)}(Tu, a) + \mathbb{M}_{2,2}^{(1,-1;a)}(Tu) \mathbb{T}_{1,1}^{(b)}(Tu, a) \right)}{6} \\
\text{z472} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(Eu) \mathbb{T}_{1,1}^{(b)}(Tu, b) + \mathbb{M}_{2,2}^{(1,-1;a)}(Eu) \mathbb{T}_{1,1}^{(b)}(Tu, b) - \mathbb{M}_{2,2}^{(1,-1;a)}(Tu) \mathbb{T}_{1,3}^{(b)}(Tu, b) + \mathbb{M}_{2,3}^{(1,-1;a)}(Tu) \mathbb{T}_{1,2}^{(b)}(Tu, b) \right)}{6} \\
\text{z473} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(Eu) \mathbb{T}_{1,2}^{(b)}(Tu, b) + \mathbb{M}_{2,1}^{(1,-1;a)}(Tu) \mathbb{T}_{1,3}^{(b)}(Tu, b) + \mathbb{M}_{2,2}^{(1,-1;a)}(Eu) \mathbb{T}_{1,2}^{(b)}(Tu, b) - \mathbb{M}_{2,3}^{(1,-1;a)}(Tu) \mathbb{T}_{1,1}^{(b)}(Tu, b) \right)}{6} \\
\text{z474} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(Tu) \mathbb{T}_{1,2}^{(b)}(Tu, b) - 2 \mathbb{M}_{2,2}^{(1,-1;a)}(Eu) \mathbb{T}_{1,3}^{(b)}(Tu, b) + \mathbb{M}_{2,2}^{(1,-1;a)}(Tu) \mathbb{T}_{1,1}^{(b)}(Tu, b) \right)}{6} \\
\text{z475} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{6} \left(\mathbb{G}_{2,2}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,3}^{(b)}(Tu, 1) + \mathbb{G}_{2,3}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,2}^{(b)}(Tu, 1) \right)}{6} \\
\text{z476} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{6} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,3}^{(b)}(Tu, 1) + \mathbb{G}_{2,3}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,1}^{(b)}(Tu, 1) \right)}{6} \\
\text{z477} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{6} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,2}^{(b)}(Tu, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,1}^{(b)}(Tu, 1) \right)}{6} \\
\text{z478} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 2) &= \frac{\sqrt{2} \left(6 \mathbb{G}_{2,1}^{(1,-1;a)}(Eu) \mathbb{Q}_{3,1}^{(b)}(Tu, 1) + 2 \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(Eu) \mathbb{Q}_{3,1}^{(b)}(Tu, 1) - \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,3}^{(b)}(Tu, 1) + \sqrt{3} \mathbb{G}_{2,3}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,2}^{(b)}(Tu, 1) \right)}{18} \\
\text{z479} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 2) &= \frac{\sqrt{2} \left(-6 \mathbb{G}_{2,1}^{(1,-1;a)}(Eu) \mathbb{Q}_{3,2}^{(b)}(Tu, 1) + \sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,3}^{(b)}(Tu, 1) + 2 \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(Eu) \mathbb{Q}_{3,2}^{(b)}(Tu, 1) - \sqrt{3} \mathbb{G}_{2,3}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,1}^{(b)}(Tu, 1) \right)}{18} \\
\text{z480} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 2) &= \frac{\sqrt{6} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,2}^{(b)}(Tu, 1) - 4 \mathbb{G}_{2,2}^{(1,-1;a)}(Eu) \mathbb{Q}_{3,3}^{(b)}(Tu, 1) + \mathbb{G}_{2,2}^{(1,-1;a)}(Tu) \mathbb{Q}_{3,1}^{(b)}(Tu, 1) \right)}{18} \\
\text{z481} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{6} \left(\mathbb{Q}_{1,2}^{(1,0;a)}(Tu) \mathbb{Q}_{1,3}^{(b)}(Tu) + \mathbb{Q}_{1,3}^{(1,0;a)}(Tu) \mathbb{Q}_{1,2}^{(b)}(Tu) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\text{z482} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z483} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z484} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, b) &= -\frac{\sqrt{6} \left(\mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{6} \\
\text{z485} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, b) &= -\frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\text{z486} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, b) &= -\frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\text{z487} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6} \\
\text{z488} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\text{z489} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\text{z490} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, d) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6} \\
\text{z491} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, d) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\text{z492} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, d) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\text{z493} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(T_g, 1) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{6} \\
\text{z494} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(T_g, 1) &= \frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z495}} \quad \mathbb{Q}_{4,3}^{(1,0;c)}(T_g, 1) = \frac{\sqrt{6} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z496}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, a) = \frac{\sqrt{6} \left(\mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) - \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z497}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, a) = \frac{\sqrt{6} \left(-\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z498}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, a) = \frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) - \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z499}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z500}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, b) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z501}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z502}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z503}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z504}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z505}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(T_u) + 3 \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,1}^{(b)}(T_u) + 3 \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) + 3 \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z506}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + 3 \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,3}^{(b)}(T_u) - 3 \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + 3 \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z507}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{10} \left(2\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{1,3}^{(b)}(T_u) + 3 \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,2}^{(b)}(T_u) + 3 \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\begin{aligned}
\boxed{\text{z508}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, b) &= \frac{-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{6} \\
\boxed{\text{z509}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, b) &= -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{6} \\
\boxed{\text{z510}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{3} \\
\boxed{\text{z511}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a)\right)}{30} \\
\boxed{\text{z512}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a)\right)}{30} \\
\boxed{\text{z513}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a)\right)}{30} \\
\boxed{\text{z514}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b)\right)}{30} \\
\boxed{\text{z515}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b)\right)}{30} \\
\boxed{\text{z516}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b)\right)}{30} \\
\boxed{\text{z517}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1a) &= \frac{\sqrt{5}\left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)\right)}{30} \\
\boxed{\text{z518}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1a) &= -\frac{\sqrt{5}\left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)\right)}{30} \\
\boxed{\text{z519}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1a) &= \frac{\sqrt{5}\left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)\right)}{15} \\
\boxed{\text{z520}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1b) &= \frac{\sqrt{5}\left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) - 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) - 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a)\right)}{30}
\end{aligned}$$

$$\begin{aligned}
\text{z521} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1b) &= -\frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{30} \\
\text{z522} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1b) &= \frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{15} \\
\text{z523} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1c) &= \frac{\sqrt{5} \left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) - 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) - 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{30} \\
\text{z524} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1c) &= -\frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{30} \\
\text{z525} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1c) &= \frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{15} \\
\text{z526} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2a) &= \frac{-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u) + 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} \\
\text{z527} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} \\
\text{z528} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2a) &= \frac{-\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} \\
\text{z529} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2b) &= \frac{-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{6} \\
\text{z530} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) - 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{6} \\
\text{z531} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2b) &= \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{3} \\
\text{z532} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2c) &= \frac{-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{6} \\
\text{z533} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2c) &= \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) - 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{6}
\end{aligned}$$

$$\boxed{\text{z534}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2c) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{3}$$

$$\boxed{\text{z535}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, a) = \frac{\sqrt{6} \left(\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u) - \mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z536}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, a) = \frac{\sqrt{6} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z537}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, a) = \frac{\sqrt{6} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z538}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z539}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, b) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z540}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, b) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z541}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z542}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, c) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z543}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z544}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z545}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z546}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z547}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{3}$$

$$\boxed{\text{z548}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{3}$$

$$\boxed{\text{z549}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,3}^{(b)}(T_u, a)}{3}$$

$$\boxed{\text{z550}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g, c) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{3}$$

$$\boxed{\text{z551}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g, c) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{3}$$

$$\boxed{\text{z552}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g, c) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,3}^{(b)}(T_u, b)}{3}$$

$$\boxed{\text{z553}} \quad \mathbb{G}_{3,1}^{(1,1;c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z554}} \quad \mathbb{G}_{3,2}^{(1,1;c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z555}} \quad \mathbb{G}_{3,3}^{(1,1;c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z925}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z926}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z927}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z928}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, b) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{30}$$

$$\begin{aligned}
\boxed{\text{z929}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, b) &= \frac{\sqrt{10} \left(3\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z930}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, b) &= \frac{\sqrt{10} \left(3\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{30} \\
\boxed{\text{z931}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, c) &= \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z932}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, c) &= \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z933}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, c) &= \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z934}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, d) &= \frac{-\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z935}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, d) &= -\frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z936}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, d) &= \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z937}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, e) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z938}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, e) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z939}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, e) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z940}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1) &= \frac{\sqrt{5} \left(-3\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) - 2\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z941}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1) &= -\frac{\sqrt{5} \left(2\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30}
\end{aligned}$$

$$\boxed{\text{z942}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1) = \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{15}$$

$$\boxed{\text{z943}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2a) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) - 2\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6}$$

$$\boxed{\text{z944}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2a) = \frac{-2\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6}$$

$$\boxed{\text{z945}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2a) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z946}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2b) = -\frac{3\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z947}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2b) = \frac{3\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z948}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2b) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z949}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{10} \left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z950}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{10} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z951}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{30} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) + 2\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z952}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, b) = \frac{3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z953}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, b) = \frac{-3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z954}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z955}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, c) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z956}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, c) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z957}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, c) = \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z958}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1a) = \frac{\sqrt{10} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - 4\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z959}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1a) = \frac{\sqrt{10} \left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - 4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z960}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1a) = \frac{\sqrt{30} \left(-2\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + 2\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z961}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1b) = \frac{\sqrt{5} \left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g) - 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) - 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z962}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1b) = -\frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z963}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1b) = \frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{15}$$

$$\boxed{\text{z964}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2a) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12}$$

$$\boxed{\text{z965}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2a) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\boxed{\text{z966}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2a) = \frac{\sqrt{6} \left(-\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z967}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2b) = \frac{-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{2,1}^{(b)}(E_g) + 3 \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z968}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2b) = -\frac{\sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{2,1}^{(b)}(E_g) + 3 \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z969}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2b) = \frac{\sqrt{3} \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z970}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2c) = \frac{-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{M}_{1,1}^{(b)}(T_g) - 2 \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) + 2 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$$\boxed{\text{z971}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2c) = \frac{\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{M}_{1,2}^{(b)}(T_g) + 2 \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{M}_{1,2}^{(b)}(T_g) - 2 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$$\boxed{\text{z972}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2c) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z973}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z974}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z975}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z976}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, b) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3 \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z977}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, b) = \frac{\sqrt{10} \left(3 \mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3} \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3 \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3 \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z978}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, b) = \frac{\sqrt{10} \left(3 \mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3 \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + 2 \sqrt{3} \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z979}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z980}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, c) = \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z981}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, c) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z982}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, d) = \frac{-\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z983}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, d) = -\frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z984}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, d) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z985}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, e) = \frac{\sqrt{6}\left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)\right)}{6}$$

$$\boxed{\text{z986}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, e) = \frac{\sqrt{6}\left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6}$$

$$\boxed{\text{z987}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, e) = \frac{\sqrt{6}\left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6}$$

$$\boxed{\text{z988}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1) = \frac{\sqrt{5}\left(-3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) - 2\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)\right)}{30}$$

$$\boxed{\text{z989}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1) = -\frac{\sqrt{5}\left(2\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)\right)}{30}$$

$$\boxed{\text{z990}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1) = \frac{\sqrt{5}\left(-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)\right)}{15}$$

$$\boxed{\text{z991}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2a) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) - 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6}$$

$$\boxed{\text{z992}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2a) = \frac{-2\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6}$$

$$\boxed{\text{z993}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2a) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z994}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2b) = -\frac{3\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z995}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2b) = \frac{3\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z996}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2b) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z997}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z998}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z999}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z1000}} \quad \mathbb{G}_{2,1}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z1001}} \quad \mathbb{G}_{2,2}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z1002}} \quad \mathbb{G}_{2,3}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(-\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g) + 2\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z1003}} \quad \mathbb{G}_{2,1}^{(c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z1004}} \quad \mathbb{G}_{2,2}^{(c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\begin{aligned}
\boxed{\text{z1005}} \quad \mathbb{G}_{2,3}^{(c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z1006}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1007}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1008}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{3} \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1009}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{14} \left(\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + \sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3 \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - 3 \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3 \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{42} \\
\boxed{\text{z1010}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{14} \left(\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3 \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3} \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3 \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{42} \\
\boxed{\text{z1011}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{14} \left(-2\sqrt{3} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3 \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) - 2\sqrt{3} \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42} \\
\boxed{\text{z1012}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1013}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1014}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{3} \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1015}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z1016}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z1017}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z1018}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 1) &= -\frac{\sqrt{2} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12} \\
\boxed{\text{z1019}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 1) &= \frac{\sqrt{2} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12} \\
\boxed{\text{z1020}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 1) &= \frac{\sqrt{6} \left(\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6} \\
\boxed{\text{z1021}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{14} \left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + 4\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{84} \\
\boxed{\text{z1022}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{14} \left(-3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) - 3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + 4\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{84} \\
\boxed{\text{z1023}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{14} \left(3\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + 2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{G}_{2,3}^{(1,-1;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42} \\
\boxed{\text{z1024}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z1025}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{2} \left(\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z1026}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{2} \left(-\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(T_g) + 2\mathbb{Q}_{1,3}^{(1,0;a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6} \\
\boxed{\text{z1027}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z1028}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z1029}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z1030}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_u) = \frac{\sqrt{3} \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z1031}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_u) = \frac{\sqrt{3} \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z1032}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_u) = \frac{\sqrt{3} \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

• **A;A_001_1** : 'A'-'A' bond-cluster

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: **12a@12j**

$$\boxed{\text{z22}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z23}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{5} \left(\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{5}$$

$$\boxed{\text{z24}} \quad \mathbb{Q}_0^{(c)}(A_g, c) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z25}} \quad \mathbb{Q}_4^{(c)}(A_g) = \frac{\sqrt{30} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z26}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g, a) = \frac{\sqrt{5} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{5}$$

$$\boxed{\text{z27}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g, b) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z28}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, a) = \frac{\sqrt{30} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{30}$$

$$\boxed{\text{z29}} \quad \mathbb{Q}_4^{(1,-1;c)}(A_g, b) = \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z30}} \quad \mathbb{Q}_0^{(1,0;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{2,1}^{(b)}(E_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z31}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g, a) = \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z32}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g, b) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z33}} \quad \mathbb{G}_3^{(c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z34}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, a) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z35}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, b) = \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_0^{(b)}(A_g)$$

$$\boxed{\text{z36}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g, c) = - \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z37}} \quad \mathbb{G}_3^{(1,0;c)}(A_g, a) = \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z38}} \quad \mathbb{G}_3^{(1,0;c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{2,2}^{(b)}(E_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z39}} \quad \mathbb{G}_3^{(1,0;c)}(A_g, c) = \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{3}$$

$$\boxed{\text{z87}} \quad \mathbb{Q}_3^{(c)}(A_u, a) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z88}} \quad \mathbb{Q}_3^{(c)}(A_u, b) = - \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3}$$

$$\boxed{\text{z89}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, a) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\begin{aligned}
\text{z90} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, b) &= -\frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3} \\
\text{z91} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, c) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3} \\
\text{z92} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u, d) &= -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3} \\
\text{z93} \quad \mathbb{Q}_3^{(1,0;c)}(A_u, a) &= \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3} \\
\text{z94} \quad \mathbb{Q}_3^{(1,0;c)}(A_u, b) &= \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3} \\
\text{z95} \quad \mathbb{G}_0^{(c)}(A_u, a) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3} \\
\text{z96} \quad \mathbb{G}_0^{(c)}(A_u, b) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3} \\
\text{z97} \quad \mathbb{G}_0^{(1,-1;c)}(A_u, a) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3} \\
\text{z98} \quad \mathbb{G}_0^{(1,-1;c)}(A_u, b) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3} \\
\text{z99} \quad \mathbb{G}_4^{(1,-1;c)}(A_u, a) &= \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3} \\
\text{z100} \quad \mathbb{G}_4^{(1,-1;c)}(A_u, b) &= \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3} \\
\text{z101} \quad \mathbb{G}_0^{(1,0;c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{3} \\
\text{z102} \quad \mathbb{G}_4^{(1,0;c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{3}
\end{aligned}$$

$$\boxed{\text{z103}} \quad \mathbb{G}_0^{(1,1;c)}(A_u, a) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{3}$$

$$\boxed{\text{z104}} \quad \mathbb{G}_0^{(1,1;c)}(A_u, b) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{3}$$

$$\boxed{\text{z177}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z178}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z179}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z180}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z181}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\sqrt{7} \left(2 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 2 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{14}$$

$$\boxed{\text{z182}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = \frac{\sqrt{7} \left(-2 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 2 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{14}$$

$$\boxed{\text{z183}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, d) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + 2 \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z184}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, d) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z185}} \quad \mathbb{Q}_{4,1}^{(c)}(E_g) = \frac{\sqrt{21} \left(3 \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 2 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 4 \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{42}$$

$$\boxed{\text{z186}} \quad \mathbb{Q}_{4,2}^{(c)}(E_g) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 2 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 2 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{14}$$

$$\boxed{\text{z187}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\begin{aligned}
\boxed{\text{z188}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2} \\
\boxed{\text{z189}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{7}\left(2\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)\right)}{14} \\
\boxed{\text{z190}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{7}\left(-2\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)\right)}{14} \\
\boxed{\text{z191}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, c) &= -\frac{\sqrt{2}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2} \\
\boxed{\text{z192}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, c) &= \frac{\sqrt{2}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} \\
\boxed{\text{z193}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, d) &= \frac{\sqrt{14}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g)\right)}{28} \\
\boxed{\text{z194}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, d) &= \frac{\sqrt{14}\left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g)\right)}{84} \\
\boxed{\text{z195}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, e) &= \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z196}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, e) &= \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{2} \\
\boxed{\text{z197}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{21}\left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 4\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)\right)}{42} \\
\boxed{\text{z198}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{7}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)\right)}{14} \\
\boxed{\text{z199}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, b) &= \frac{\sqrt{14}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g)\right)}{84}
\end{aligned}$$

$$\boxed{\text{z200}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{14} \left(\sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) + \sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) - 2\sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{28}$$

$$\boxed{\text{z201}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z202}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, a) = \frac{\sqrt{3} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z203}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, b) = \frac{\sqrt{2} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z204}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, b) = \frac{\sqrt{2} \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z205}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, c) = \frac{\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{2,1}^{(b)}(E_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z206}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, c) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{2,2}^{(b)}(E_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z207}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g, d) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z208}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g, d) = \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z209}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z210}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z211}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g, b) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z212}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g, b) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$$\boxed{\text{z307}} \quad \mathbb{Q}_{5,1}^{(c)}(E_u) = \frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$$

$$\boxed{\text{z308}} \quad \mathbb{Q}_{5,2}^{(c)}(E_u) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z309}} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(E_u) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$$

$$\boxed{\text{z310}} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{3} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z311}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, a) = \frac{-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z312}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, a) = \frac{\sqrt{3} \left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z313}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, b) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z314}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$$

$$\boxed{\text{z315}} \quad \mathbb{G}_{2,1}^{(c)}(E_u, c) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z316}} \quad \mathbb{G}_{2,2}^{(c)}(E_u, c) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$$

$$\boxed{\text{z317}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z318}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{3} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z319}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, a) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{28}$$

$$\boxed{\text{z320}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{14} \left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, a) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{84}$$

$$\boxed{\text{z321}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, c) = \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, b) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{28}$$

$$\boxed{\text{z322}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, c) = \frac{\sqrt{14} \left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, b) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{84}$$

$$\boxed{\text{z323}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, d) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z324}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, d) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$$

$$\boxed{\text{z325}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, e) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z326}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, e) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$$

$$\boxed{\text{z327}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, a) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, a) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{84}$$

$$\boxed{\text{z328}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14} \left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, a) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, a) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, a) - 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{28}$$

$$\boxed{\text{z329}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, b) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, b) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{84}$$

$$\boxed{\text{z330}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{14} \left(\sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, b) - 2\sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{28}$$

$$\boxed{\text{z331}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, a) = \frac{\sqrt{3} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z332}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z333}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, b) = \frac{\sqrt{3} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) \right)}{6}$$

$$\boxed{\text{z334}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, b) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$$

$$\boxed{\text{z335}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, c) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$$

$$\boxed{\text{z336}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, c) = \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z337}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u, d) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$$

$$\boxed{\text{z338}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u, d) = \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z339}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, a) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z340}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, a) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$$

$$\boxed{\text{z341}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u, b) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z342}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$$

$$\boxed{\text{z556}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z557}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z558}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z559}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z560}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z561}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z562}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, c) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{42}$$

$$\boxed{\text{z563}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, c) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{42}$$

$$\boxed{\text{z564}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, c) = \frac{\sqrt{14} \left(-2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42}$$

$$\boxed{\text{z565}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, d) = -\frac{3\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z566}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, d) = \frac{3\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z567}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, d) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z568}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, e) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z569}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, e) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z570}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, e) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z571}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 1) = -\frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12}$$

$$\boxed{\text{z572}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\boxed{\text{z573}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{6} \left(\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z574}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 2) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{84}$$

$$\boxed{\text{z575}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 2) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{84}$$

$$\boxed{\text{z576}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 2) = \frac{\sqrt{14} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42}$$

$$\boxed{\text{z577}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z578}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z579}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z580}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{14} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{42}$$

$$\begin{aligned}
\boxed{\text{z581}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{14} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3 \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3} \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3 \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{42} \\
\boxed{\text{z582}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{14} \left(-2\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3 \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 2\sqrt{3} \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42} \\
\boxed{\text{z583}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(E_g) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(E_g) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(E_g) \right)}{12} \\
\boxed{\text{z584}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,1}^{(b)}(E_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(E_g) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(E_g) \right)}{12} \\
\boxed{\text{z585}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, c) &= -\frac{\sqrt{2} \left(\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{2,2}^{(b)}(E_g) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(E_g) \right)}{6} \\
\boxed{\text{z586}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{21} \\
\boxed{\text{z587}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,2}^{(b)}(T_g) \right)}{21} \\
\boxed{\text{z588}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,3}^{(b)}(T_g) \right)}{21} \\
\boxed{\text{z589}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, e) &= -\frac{3 \mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z590}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, e) &= \frac{3 \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3} \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z591}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, e) &= \frac{\sqrt{3} \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z592}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, f) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z593}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, f) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z594}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, f) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z595}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1a) = -\frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12}$$

$$\boxed{\text{z596}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1a) = \frac{\sqrt{2} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\boxed{\text{z597}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1a) = \frac{\sqrt{6} \left(\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z598}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1b) = -\frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z599}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1b) = \frac{3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z600}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1b) = \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z601}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1c) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g)}{12}$$

$$\boxed{\text{z602}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1c) = \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g)}{12}$$

$$\boxed{\text{z603}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1c) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g)}{12}$$

$$\boxed{\text{z604}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 2a) = \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{84}$$

$$\begin{aligned}
\text{z605} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{14} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{84} \\
\text{z606} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{14} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{42} \\
\text{z607} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{2} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(E_g) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(E_g) \right)}{12} \\
\text{z608} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{2} \left(\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(E_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,2}^{(b)}(E_g) \right)}{12} \\
\text{z609} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{2} \left(\sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(E_g) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{2,1}^{(b)}(E_g) \right)}{6} \\
\text{z610} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 2c) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) + 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{84} \\
\text{z611} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 2c) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) - 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{84} \\
\text{z612} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 2c) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g) - 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g) + 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{M}_{1,3}^{(b)}(T_g) \right)}{84} \\
\text{z613} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{6} \\
\text{z614} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{2} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{6} \\
\text{z615} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{2} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{6} \\
\text{z616} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}
\end{aligned}$$

$$\boxed{\text{z617}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z618}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z619}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, c) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) - 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z620}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, c) = \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z621}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, c) = -\frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z622}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g, d) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z623}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g, d) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z624}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g, d) = \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - 2\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z625}} \quad \mathbb{Q}_{4,1}^{(1,0;c)}(T_g, 1) = -\frac{3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z626}} \quad \mathbb{Q}_{4,2}^{(1,0;c)}(T_g, 1) = \frac{3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z627}} \quad \mathbb{Q}_{4,3}^{(1,0;c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z628}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z629}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$$\begin{aligned}
\boxed{\text{z630}} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3} \\
\boxed{\text{z631}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_g, b) &= -\frac{3\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z632}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_g, b) &= \frac{3\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z633}} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z634}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(T_g, c) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z635}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(T_g, c) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z636}} \quad \mathbb{Q}_{2,3}^{(1,1;c)}(T_g, c) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)\right)}{6} \\
\boxed{\text{z637}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, a) &= \frac{\sqrt{10}\left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z638}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, a) &= \frac{\sqrt{10}\left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)\right)}{30} \\
\boxed{\text{z639}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, a) &= \frac{\sqrt{30}\left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 2\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)\right)}{30} \\
\boxed{\text{z640}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z641}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z642}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}
\end{aligned}$$

$$\boxed{\text{z643}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, c) = \frac{-\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z644}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, c) = -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z645}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, c) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z646}} \quad \mathbb{G}_{1,1}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z647}} \quad \mathbb{G}_{1,2}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z648}} \quad \mathbb{G}_{1,3}^{(c)}(T_g, d) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z649}} \quad \mathbb{G}_{3,1}^{(c)}(T_g, 1) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z650}} \quad \mathbb{G}_{3,2}^{(c)}(T_g, 1) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z651}} \quad \mathbb{G}_{3,3}^{(c)}(T_g, 1) = \frac{\sqrt{30} \left(-2\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 2\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z652}} \quad \mathbb{G}_{3,1}^{(c)}(T_g, 2) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12}$$

$$\boxed{\text{z653}} \quad \mathbb{G}_{3,2}^{(c)}(T_g, 2) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\boxed{\text{z654}} \quad \mathbb{G}_{3,3}^{(c)}(T_g, 2) = \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{6}$$

$$\boxed{\text{z655}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{30}$$

$$\begin{aligned}
\boxed{\text{z656}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, a) &= \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z657}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, a) &= \frac{\sqrt{30} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) + 2\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30} \\
\boxed{\text{z658}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, b) &= \frac{-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z659}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, b) &= -\frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z660}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z661}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z662}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z663}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, c) &= \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z664}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, d) &= \frac{-\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z665}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, d) &= -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z666}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, d) &= \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z667}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g, e) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6} \\
\boxed{\text{z668}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g, e) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z669}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g, e) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z670}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1a) = \frac{\sqrt{10} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z671}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1a) = \frac{\sqrt{10} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{60}$$

$$\boxed{\text{z672}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1a) = \frac{\sqrt{30} \left(-2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{30}$$

$$\boxed{\text{z673}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1b) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z674}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1b) = \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z675}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1b) = \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z676}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1c) = \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g)}{12}$$

$$\boxed{\text{z677}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1c) = \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g)}{12}$$

$$\boxed{\text{z678}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1c) = \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g)}{12}$$

$$\boxed{\text{z679}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2a) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{12}$$

$$\boxed{\text{z680}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2a) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{12}$$

$$\begin{aligned}
\text{z681} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2a) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{6} \\
\text{z682} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_0^{(b)}(A_g)}{3} \\
\text{z683} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_0^{(b)}(A_g)}{3} \\
\text{z684} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2b) &= \frac{\sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_0^{(b)}(A_g)}{3} \\
\text{z685} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2c) &= \frac{\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) + \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) + 4 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,1}^{(b)}(T_g)}{12} \\
\text{z686} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2c) &= \frac{\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,3}^{(b)}(T_g) + \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) + 4 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,2}^{(b)}(T_g)}{12} \\
\text{z687} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2c) &= \frac{\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,2}^{(b)}(T_g) + \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{M}_{1,1}^{(b)}(T_g) + 4 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{M}_{1,3}^{(b)}(T_g)}{12} \\
\text{z688} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3} \mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3} \\
\text{z689} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3} \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3} \\
\text{z690} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3} \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3} \\
\text{z691} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3 \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) + 3 \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{30} \\
\text{z692} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{10} \left(3 \mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,3}^{(b)}(T_g) - \sqrt{3} \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3 \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + 3 \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30} \\
\text{z693} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, b) &= \frac{\sqrt{10} \left(3 \mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(T_g) + 3 \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(T_g) + 2 \sqrt{3} \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{30}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z694}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z695}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z696}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, c) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z697}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_g, 1a) &= \frac{\sqrt{5} \left(-3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - 2\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z698}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_g, 1a) &= -\frac{\sqrt{5} \left(2\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z699}} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_g, 1a) &= \frac{\sqrt{5} \left(-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + 3\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{15} \\
\boxed{\text{z700}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_g, 1b) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z701}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_g, 1b) &= -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{30} \\
\boxed{\text{z702}} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_g, 1b) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{15} \\
\boxed{\text{z703}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_g, 2a) &= \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) - 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6} \\
\boxed{\text{z704}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_g, 2a) &= \frac{-2\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g) + \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + 2\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6} \\
\boxed{\text{z705}} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_g, 2a) &= \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g) - \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z706}} \quad \mathbb{G}_{3,1}^{(1,0;c)}(T_g, 2b) &= \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}
\end{aligned}$$

$$\boxed{\text{z707}} \quad \mathbb{G}_{3,2}^{(1,0;c)}(T_g, 2b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$$\boxed{\text{z708}} \quad \mathbb{G}_{3,3}^{(1,0;c)}(T_g, 2b) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3}$$

$$\boxed{\text{z709}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z710}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z711}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z712}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g, b) = \frac{-\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z713}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g) + 3\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z714}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z715}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z716}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g, c) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z717}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g) \right)}{6}$$

$$\boxed{\text{z1033}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z1034}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$$\begin{aligned}
\boxed{\text{z1035}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} \\
\boxed{\text{z1036}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, b) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)\right)}{30} \\
\boxed{\text{z1037}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, b) &= \frac{\sqrt{10}\left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)\right)}{30} \\
\boxed{\text{z1038}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, b) &= \frac{\sqrt{10}\left(2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)\right)}{30} \\
\boxed{\text{z1039}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, c) &= \frac{\sqrt{5}\left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)\right)}{30} \\
\boxed{\text{z1040}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, c) &= -\frac{\sqrt{5}\left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)\right)}{30} \\
\boxed{\text{z1041}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, c) &= \frac{\sqrt{5}\left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)\right)}{15} \\
\boxed{\text{z1042}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, d) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)\right)}{6} \\
\boxed{\text{z1043}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, d) &= \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)\right)}{6} \\
\boxed{\text{z1044}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, d) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)\right)}{6} \\
\boxed{\text{z1045}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, e) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b)\right)}{6} \\
\boxed{\text{z1046}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, e) &= \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b)\right)}{6} \\
\boxed{\text{z1047}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, e) &= \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b)\right)}{6}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z1048}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{3} \\
\boxed{\text{z1049}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{3} \\
\boxed{\text{z1050}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{3} \\
\boxed{\text{z1051}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1b) &= \frac{\sqrt{5} \left(-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z1052}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1b) &= -\frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z1053}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1b) &= \frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{15} \\
\boxed{\text{z1054}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 1c) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{30} \\
\boxed{\text{z1055}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 1c) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 3\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\boxed{\text{z1056}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 1c) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\boxed{\text{z1057}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2a) &= \frac{-\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} \\
\boxed{\text{z1058}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2a) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - 2\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} \\
\boxed{\text{z1059}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2a) &= \frac{-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} \\
\boxed{\text{z1060}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2b) &= \frac{\sqrt{2} \left(6\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{18}
\end{aligned}$$

$$\begin{aligned}
\text{z1061} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2b) &= \frac{\sqrt{2} \left(-6\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{18} \\
\text{z1062} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2b) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 4\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{18} \\
\text{z1063} \quad \mathbb{Q}_{5,1}^{(c)}(T_u, 3) &= \frac{3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{18} \\
\text{z1064} \quad \mathbb{Q}_{5,2}^{(c)}(T_u, 3) &= \frac{-3\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 4\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{18} \\
\text{z1065} \quad \mathbb{Q}_{5,3}^{(c)}(T_u, 3) &= \frac{\sqrt{3} \left(2\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 2\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{9} \\
\text{z1066} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{30} \\
\text{z1067} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z1068} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z1069} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{5} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{30} \\
\text{z1070} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, b) &= -\frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\text{z1071} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{15} \\
\text{z1072} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6} \\
\text{z1073} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\text{z1074} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, c) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\text{z1075} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6} \\
\text{z1076} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\text{z1077} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\text{z1078} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1a) &= \frac{\sqrt{5} \left(-3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{30} \\
\text{z1079} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1a) &= -\frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z1080} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1a) &= \frac{\sqrt{5} \left(3\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{15} \\
\text{z1081} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{30} \\
\text{z1082} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) - 3\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\text{z1083} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1b) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{30} \\
\text{z1084} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1c) &= \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a)}{12} \\
\text{z1085} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1c) &= \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a)}{12} \\
\text{z1086} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1c) &= \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a)}{12}
\end{aligned}$$

$$\begin{aligned}
\text{z1087} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1d) &= \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u, b) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{12} \\
\text{z1088} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1d) &= \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u, b) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{12} \\
\text{z1089} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1d) &= \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{12} \\
\text{z1090} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2a) &= \frac{-\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} \\
\text{z1091} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2a) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + 2\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u) - 2\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} \\
\text{z1092} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2a) &= \frac{-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} \\
\text{z1093} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2b) &= \frac{\sqrt{2} \left(6\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{18} \\
\text{z1094} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2b) &= \frac{\sqrt{2} \left(-6\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + 2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - \sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{18} \\
\text{z1095} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2b) &= \frac{\sqrt{6} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) - 4\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{18} \\
\text{z1096} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2c) &= \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, a) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{12} \\
\text{z1097} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2c) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, a) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{12} \\
\text{z1098} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2c) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, a) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u, a) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u, a)}{12} \\
\text{z1099} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2d) &= \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u, b) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u, b) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{12}
\end{aligned}$$

$$\begin{aligned}
\text{z1100} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(Tu, 2d) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(Tg, 1)\mathbb{T}_{1,3}^{(b)}(Tu, b) - \mathbb{M}_{3,1}^{(1,-1;a)}(Tg, 2)\mathbb{T}_{1,3}^{(b)}(Tu, b) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(Tg, 1)\mathbb{T}_{1,1}^{(b)}(Tu, b) + \mathbb{M}_{3,3}^{(1,-1;a)}(Tg, 2)\mathbb{T}_{1,1}^{(b)}(Tu, b) + 4\mathbb{M}_3^{(1,-1;a)}(Ag)\mathbb{T}_{1,2}^{(b)}(Tu, b)}{12} \\
\text{z1101} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(Tu, 2d) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(Tg, 1)\mathbb{T}_{1,2}^{(b)}(Tu, b) + \mathbb{M}_{3,1}^{(1,-1;a)}(Tg, 2)\mathbb{T}_{1,2}^{(b)}(Tu, b) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(Tg, 1)\mathbb{T}_{1,1}^{(b)}(Tu, b) - \mathbb{M}_{3,2}^{(1,-1;a)}(Tg, 2)\mathbb{T}_{1,1}^{(b)}(Tu, b) + 4\mathbb{M}_3^{(1,-1;a)}(Ag)\mathbb{T}_{1,3}^{(b)}(Tu, b)}{12} \\
\text{z1102} \quad \mathbb{Q}_{5,1}^{(1,-1;c)}(Tu, 3) &= \frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(Eg)\mathbb{Q}_{3,1}^{(b)}(Tu, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(Eg)\mathbb{Q}_{3,1}^{(b)}(Tu, 1) + 4\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(Tg)\mathbb{Q}_{3,3}^{(b)}(Tu, 1) - 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(Tg)\mathbb{Q}_{3,2}^{(b)}(Tu, 1)}{18} \\
\text{z1103} \quad \mathbb{Q}_{5,2}^{(1,-1;c)}(Tu, 3) &= \frac{-3\mathbb{Q}_{2,1}^{(1,-1;a)}(Eg)\mathbb{Q}_{3,2}^{(b)}(Tu, 1) - 4\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(Tg)\mathbb{Q}_{3,3}^{(b)}(Tu, 1) + \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(Eg)\mathbb{Q}_{3,2}^{(b)}(Tu, 1) + 4\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(Tg)\mathbb{Q}_{3,1}^{(b)}(Tu, 1)}{18} \\
\text{z1104} \quad \mathbb{Q}_{5,3}^{(1,-1;c)}(Tu, 3) &= \frac{\sqrt{3} \left(2\mathbb{Q}_{2,1}^{(1,-1;a)}(Tg)\mathbb{Q}_{3,2}^{(b)}(Tu, 1) - \mathbb{Q}_{2,2}^{(1,-1;a)}(Eg)\mathbb{Q}_{3,3}^{(b)}(Tu, 1) - 2\mathbb{Q}_{2,2}^{(1,-1;a)}(Tg)\mathbb{Q}_{3,1}^{(b)}(Tu, 1) \right)}{9} \\
\text{z1105} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(Tu, a) &= \frac{\sqrt{6} \left(\mathbb{G}_{1,2}^{(1,0;a)}(Tg)\mathbb{Q}_{1,3}^{(b)}(Tu) - \mathbb{G}_{1,3}^{(1,0;a)}(Tg)\mathbb{Q}_{1,2}^{(b)}(Tu) \right)}{6} \\
\text{z1106} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(Tu, a) &= \frac{\sqrt{6} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(Tg)\mathbb{Q}_{1,3}^{(b)}(Tu) + \mathbb{G}_{1,3}^{(1,0;a)}(Tg)\mathbb{Q}_{1,1}^{(b)}(Tu) \right)}{6} \\
\text{z1107} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(Tu, a) &= \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(Tg)\mathbb{Q}_{1,2}^{(b)}(Tu) - \mathbb{G}_{1,2}^{(1,0;a)}(Tg)\mathbb{Q}_{1,1}^{(b)}(Tu) \right)}{6} \\
\text{z1108} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(Tu, b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(Eg)\mathbb{T}_{1,1}^{(b)}(Tu, a) + 3\mathbb{T}_{2,2}^{(1,0;a)}(Eg)\mathbb{T}_{1,1}^{(b)}(Tu, a) + 3\mathbb{T}_{2,2}^{(1,0;a)}(Tg)\mathbb{T}_{1,3}^{(b)}(Tu, a) + 3\mathbb{T}_{2,3}^{(1,0;a)}(Tg)\mathbb{T}_{1,2}^{(b)}(Tu, a) \right)}{30} \\
\text{z1109} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(Tu, b) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(Eg)\mathbb{T}_{1,2}^{(b)}(Tu, a) + 3\mathbb{T}_{2,1}^{(1,0;a)}(Tg)\mathbb{T}_{1,3}^{(b)}(Tu, a) - 3\mathbb{T}_{2,2}^{(1,0;a)}(Eg)\mathbb{T}_{1,2}^{(b)}(Tu, a) + 3\mathbb{T}_{2,3}^{(1,0;a)}(Tg)\mathbb{T}_{1,1}^{(b)}(Tu, a) \right)}{30} \\
\text{z1110} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(Tu, b) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(Eg)\mathbb{T}_{1,3}^{(b)}(Tu, a) + 3\mathbb{T}_{2,1}^{(1,0;a)}(Tg)\mathbb{T}_{1,2}^{(b)}(Tu, a) + 3\mathbb{T}_{2,2}^{(1,0;a)}(Tg)\mathbb{T}_{1,1}^{(b)}(Tu, a) \right)}{30} \\
\text{z1111} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(Tu, c) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(Eg)\mathbb{T}_{1,1}^{(b)}(Tu, b) + 3\mathbb{T}_{2,2}^{(1,0;a)}(Eg)\mathbb{T}_{1,1}^{(b)}(Tu, b) + 3\mathbb{T}_{2,2}^{(1,0;a)}(Tg)\mathbb{T}_{1,3}^{(b)}(Tu, b) + 3\mathbb{T}_{2,3}^{(1,0;a)}(Tg)\mathbb{T}_{1,2}^{(b)}(Tu, b) \right)}{30} \\
\text{z1112} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(Tu, c) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(Eg)\mathbb{T}_{1,2}^{(b)}(Tu, b) + 3\mathbb{T}_{2,1}^{(1,0;a)}(Tg)\mathbb{T}_{1,3}^{(b)}(Tu, b) - 3\mathbb{T}_{2,2}^{(1,0;a)}(Eg)\mathbb{T}_{1,2}^{(b)}(Tu, b) + 3\mathbb{T}_{2,3}^{(1,0;a)}(Tg)\mathbb{T}_{1,1}^{(b)}(Tu, b) \right)}{30}
\end{aligned}$$

$$\begin{aligned}
\text{z1113} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, c) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{30} \\
\text{z1114} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1a) &= \frac{\sqrt{6} \left(-\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{6} \\
\text{z1115} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1a) &= \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) - \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\text{z1116} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1a) &= \frac{\sqrt{6} \left(-\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\text{z1117} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1b) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{30} \\
\text{z1118} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1b) &= -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{30} \\
\text{z1119} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1b) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{15} \\
\text{z1120} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1c) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{30} \\
\text{z1121} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1c) &= -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{30} \\
\text{z1122} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1c) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{15} \\
\text{z1123} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2a) &= -\frac{\sqrt{6} \left(\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) \right)}{6} \\
\text{z1124} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2a) &= -\frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u, 1) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6} \\
\text{z1125} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2a) &= -\frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u, 1) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u, 1) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\boxed{\text{z1126}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2b) &= \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{6} \\
\boxed{\text{z1127}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2b) &= \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{6} \\
\boxed{\text{z1128}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2b) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{3} \\
\boxed{\text{z1129}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2c) &= \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u, b) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{6} \\
\boxed{\text{z1130}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2c) &= \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{6} \\
\boxed{\text{z1131}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2c) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{3} \\
\boxed{\text{z1132}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} \\
\boxed{\text{z1133}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} \\
\boxed{\text{z1134}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} \\
\boxed{\text{z1135}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1136}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u, b) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1137}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1138}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u, c) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z1139}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u, c) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z1140}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z1141}} \quad \mathbb{Q}_{3,1}^{(1,1;c)}(T_u, 1) = \frac{\sqrt{3} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z1142}} \quad \mathbb{Q}_{3,2}^{(1,1;c)}(T_u, 1) = \frac{\sqrt{3} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z1143}} \quad \mathbb{Q}_{3,3}^{(1,1;c)}(T_u, 1) = \frac{\sqrt{3} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{3}$$

$$\boxed{\text{z1144}} \quad \mathbb{G}_{2,1}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1145}} \quad \mathbb{G}_{2,2}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(T_u) - \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1146}} \quad \mathbb{G}_{2,3}^{(c)}(T_u, a) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) - 2 \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1147}} \quad \mathbb{G}_{2,1}^{(c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z1148}} \quad \mathbb{G}_{2,2}^{(c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z1149}} \quad \mathbb{G}_{2,3}^{(c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z1150}} \quad \mathbb{G}_{2,1}^{(c)}(T_u, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z1151}} \quad \mathbb{G}_{2,2}^{(c)}(T_u, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z1152}} \quad \mathbb{G}_{2,3}^{(c)}(T_u, c) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}$$

$$\boxed{\text{z1153}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,1}^{(b)}(T_u) - \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1154}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,2}^{(b)}(T_u) - \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1155}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{2} \left(-\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) - 2 \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1156}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{21}$$

$$\boxed{\text{z1157}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{21}$$

$$\boxed{\text{z1158}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{21}$$

$$\boxed{\text{z1159}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, c) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{21}$$

$$\boxed{\text{z1160}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, c) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{21}$$

$$\boxed{\text{z1161}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, c) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{21}$$

$$\begin{aligned}
\boxed{\text{z1162}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1163}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1164}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, d) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1165}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, e) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z1166}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, e) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z1167}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, e) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z1168}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 1a) &= \frac{\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a)}{12} \\
\boxed{\text{z1169}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 1a) &= \frac{-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a)}{12} \\
\boxed{\text{z1170}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 1a) &= \frac{\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a)}{12} \\
\boxed{\text{z1171}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 1b) &= \frac{\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, b) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, b) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, b)}{12} \\
\boxed{\text{z1172}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 1b) &= \frac{-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, b) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, b) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, b)}{12} \\
\boxed{\text{z1173}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 1b) &= \frac{\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, b) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, b) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, b)}{12} \\
\boxed{\text{z1174}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 2a) &= \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) - 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) + 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{84}
\end{aligned}$$

$$\boxed{\text{z1175}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 2a) \\ = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, a) + 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, a) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) - 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{84}$$

$$\boxed{\text{z1176}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 2a) \\ = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, a) - 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, a) + 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, a) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) \right)}{84}$$

$$\boxed{\text{z1177}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 2b) \\ = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, b) - 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, b) + 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, b) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{84}$$

$$\boxed{\text{z1178}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 2b) \\ = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u, b) + 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u, b) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, b) - 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, b) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{84}$$

$$\boxed{\text{z1179}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 2b) \\ = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u, b) - 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u, b) + 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u, b) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) \right)}{84}$$

$$\boxed{\text{z1180}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1181}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,3}^{(b)}(T_u) + \mathbb{G}_{1,3}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1182}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u, a) = \frac{\sqrt{6} \left(\mathbb{G}_{1,1}^{(1,0;a)}(T_g) \mathbb{Q}_{1,2}^{(b)}(T_u) + \mathbb{G}_{1,2}^{(1,0;a)}(T_g) \mathbb{Q}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1183}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u, b) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6}$$

$$\boxed{\text{z1184}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u, b) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6}$$

$$\begin{aligned}
\boxed{\text{z1185}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u, b) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) - 2\mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1186}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u, c) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z1187}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u, c) &= \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z1188}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u, c) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) - 2\mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z1189}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1190}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, a) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1191}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_u, a) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, a) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, a) \right)}{6} \\
\boxed{\text{z1192}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z1193}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u, b) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6} \\
\boxed{\text{z1194}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u, b) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u, b) \right)}{6}
\end{aligned}$$

• **A;A_002_1** : 'A'-'A' bond-cluster

- * bra: $\langle s, \uparrow |, \langle s, \downarrow |$
- * ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$
- * wyckoff: **3a@1a**

$$\boxed{\text{z40}} \quad \mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z105}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z213}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z214}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z343}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z344}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z1195}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1196}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1197}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1198}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1199}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1200}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

• **A;A_002_1** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |$, $\langle s, \downarrow |$

* ket: $|p_x, \uparrow\rangle$, $|p_x, \downarrow\rangle$, $|p_y, \uparrow\rangle$, $|p_y, \downarrow\rangle$, $|p_z, \uparrow\rangle$, $|p_z, \downarrow\rangle$

* wyckoff: **3a@1a**

$$\boxed{\text{z41}} \quad \mathbb{Q}_0^{(c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z42}} \quad \mathbb{Q}_0^{(1,0;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z43}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z106}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z107}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z108}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z215}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z216}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z217}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z218}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z219}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z220}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z345}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z346}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z347}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z348}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z349}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z350}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z718}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z719}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z720}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z721}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z722}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z723}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z724}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z725}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z726}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\begin{aligned}
\boxed{\text{z727}} \quad \mathbb{G}_{1,1}^{(c)}(T_g) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z728}} \quad \mathbb{G}_{1,2}^{(c)}(T_g) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z729}} \quad \mathbb{G}_{1,3}^{(c)}(T_g) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z730}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g) &= \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) + 3 \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) + 3 \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + 3 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z731}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g) &= \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 3 \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - 3 \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 3 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z732}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g) &= \frac{\sqrt{10} \left(2\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,3}^{(b)}(T_u) + 3 \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 3 \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z733}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{5} \left(-3 \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) + 3\sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) - 2\sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - 2\sqrt{3} \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z734}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1) &= -\frac{\sqrt{5} \left(3 \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + 3\sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3} \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z735}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{5} \left(3 \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{15} \\
\boxed{\text{z736}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2) &= \frac{-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) - 2 \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + 2 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{6} \\
\boxed{\text{z737}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2) &= \frac{\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2 \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) - 2 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u)}{6} \\
\boxed{\text{z738}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2) &= \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u)}{3} \\
\boxed{\text{z739}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z740}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z741}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z742}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g) = \frac{\sqrt{3} \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z743}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g) = \frac{\sqrt{3} \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z744}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g) = \frac{\sqrt{3} \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z1201}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1202}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1203}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1204}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, b) = \frac{-\sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1205}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, b) = -\frac{\sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1206}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{3} \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1207}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2) = -\frac{3 \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1208}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2) = \frac{3 \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\begin{aligned}
\boxed{\text{z1209}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z1210}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) &= \frac{3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1211}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) &= \frac{-3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1212}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) &= -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z1213}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) &= \frac{-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1214}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) &= -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1215}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z1216}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1217}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1218}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1219}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, b) &= \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1220}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, b) &= -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1221}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, b) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z1222}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) &= -\frac{3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}
\end{aligned}$$

$$\boxed{\text{z1223}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) = \frac{3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1224}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1225}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1226}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1227}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

• **A;A_002_1 : 'A'-'A' bond-cluster**

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: **3a@1a**

$$\boxed{\text{z44}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z45}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z46}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z47}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g) = \mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z48}} \quad \mathbb{G}_3^{(c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z49}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z109}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u) = - \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\begin{aligned}
\text{z110} \quad \mathbb{Q}_3^{(1,0;c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\text{z111} \quad \mathbb{G}_0^{(c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\text{z112} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\text{z113} \quad \mathbb{G}_4^{(1,-1;c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\text{z114} \quad \mathbb{G}_0^{(1,1;c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\text{z221} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2} \\
\text{z222} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2} \\
\text{z223} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) &= \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2} \\
\text{z224} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) &= \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2} \\
\text{z225} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) &= \frac{\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2} \\
\text{z226} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) &= -\frac{\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2} \\
\text{z227} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2} \\
\text{z228} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) &= \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}
\end{aligned}$$

$$\boxed{\text{z229}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z230}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z231}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z232}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z351}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z352}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z353}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{28}$$

$$\boxed{\text{z354}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14} \left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84}$$

$$\boxed{\text{z355}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z356}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z357}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84}$$

$$\boxed{\text{z358}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{14} \left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{28}$$

$$\boxed{\text{z359}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z360}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z361}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z362}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z745}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z746}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z747}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z748}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, b) = \frac{\sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - 3 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z749}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, b) = \frac{\sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z750}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, b) = -\frac{\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z751}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 1) = -\frac{3 \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z752}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{3 \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{2,2}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z753}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{3} \mathbb{Q}_{2,3}^{(a)}(T_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z754}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3} \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\begin{aligned}
\boxed{\text{z755}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z756}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z757}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z758}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z759}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) &= -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z760}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1) &= -\frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z761}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1) &= \frac{3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z762}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z763}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g) &= -\frac{3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z764}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g) &= \frac{3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z765}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z766}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z767}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z768}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}
\end{aligned}$$

$$\boxed{\text{z769}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, b) = \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z770}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z771}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1228}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1229}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1230}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1231}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1232}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1233}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1234}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1) = \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1235}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1) = \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1236}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1) = \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1237}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1238}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1239}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1240}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z1241}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z1242}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u) = \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z1243}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1) = \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z1244}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1) = -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z1245}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1) = \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{15}$$

$$\boxed{\text{z1246}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) = \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$$\boxed{\text{z1247}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$$\boxed{\text{z1248}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z1249}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1250}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1251}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1252}} \quad \mathbb{G}_{2,1}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1253}} \quad \mathbb{G}_{2,2}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1254}} \quad \mathbb{G}_{2,3}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1255}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{21}$$

$$\boxed{\text{z1256}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{21}$$

$$\boxed{\text{z1257}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{21}$$

$$\boxed{\text{z1258}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1259}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1260}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1261}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 1) = \frac{\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1262}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 1) = \frac{-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1263}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 1) = \frac{\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\begin{aligned}
\boxed{\text{z1264}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{84} \\
\boxed{\text{z1265}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) + 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{84} \\
\boxed{\text{z1266}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84} \\
\boxed{\text{z1267}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u) &= \frac{\sqrt{2} \left(\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z1268}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u) &= \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z1269}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u) &= \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) - 2 \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z1270}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z1271}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z1272}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}
\end{aligned}$$

• **A;A_002_2** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$
* ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$
* wyckoff: **3b@3c**

$$\begin{aligned}
\boxed{\text{z50}} \quad \mathbb{Q}_0^{(c)}(A_g) &= \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_0^{(b)}(A_g) \\
\boxed{\text{z115}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) &= \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3} \\
\boxed{\text{z233}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) &= \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}
\end{aligned}$$

$$\boxed{\text{z234}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z363}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z364}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z1273}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1274}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1275}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1276}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1277}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1278}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

• **A;A_002_2** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |$, $\langle s, \downarrow |$

* ket: $|p_x, \uparrow\rangle$, $|p_x, \downarrow\rangle$, $|p_y, \uparrow\rangle$, $|p_y, \downarrow\rangle$, $|p_z, \uparrow\rangle$, $|p_z, \downarrow\rangle$

* wyckoff: **3b03c**

$$\boxed{\text{z51}} \quad \mathbb{Q}_0^{(c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z52}} \quad \mathbb{Q}_0^{(1,0;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z53}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z116}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z117}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z118}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z235}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z236}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z237}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z238}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z239}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z240}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z365}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z366}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z367}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z368}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z369}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z370}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z772}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z773}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z774}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z775}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z776}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z777}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z778}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z779}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z780}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z781}} \quad \mathbb{G}_{1,1}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z782}} \quad \mathbb{G}_{1,2}^{(c)}(T_g) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z783}} \quad \mathbb{G}_{1,3}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z784}} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) + 3 \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) + 3 \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + 3 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z785}} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g) = \frac{\sqrt{10} \left(-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 3 \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - 3 \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 3 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z786}} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g) = \frac{\sqrt{10} \left(2\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,3}^{(b)}(T_u) + 3 \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 3 \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z787}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1) = \frac{\sqrt{5} \left(-3 \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) + 3\sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) - 2\sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - 2\sqrt{3} \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z788}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1) = -\frac{\sqrt{5} \left(3 \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + 3\sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3} \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z789}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1) = \frac{\sqrt{5} \left(3 \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{15}$$

$$\boxed{\text{z790}} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2) = \frac{-\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,1}^{(b)}(T_u) - 2 \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + 2 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$$\boxed{\text{z791}} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2) = \frac{\sqrt{3} \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2 \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,2}^{(b)}(T_u) - 2 \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$$\boxed{\text{z792}} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z793}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z794}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z795}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z796}} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g) = \frac{\sqrt{3} \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z797}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g) = \frac{\sqrt{3} \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z798}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g) = \frac{\sqrt{3} \mathbb{M}_0^{(1,1;a)}(A_u) \mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z1279}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1280}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1281}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3} \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1282}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, b) = \frac{-\sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1283}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, b) = -\frac{\sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + 3 \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1284}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{3} \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1285}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2) = -\frac{3 \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3} \mathbb{Q}_{1,1}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1286}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2) = \frac{3 \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3} \mathbb{Q}_{1,2}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1287}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2) = \frac{\sqrt{3} \mathbb{Q}_{1,3}^{(a)}(T_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\begin{aligned}
\boxed{\text{z1288}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) &= \frac{3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1289}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) &= \frac{-3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1290}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) &= -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z1291}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) &= \frac{-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1292}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) &= -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1293}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z1294}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1295}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1296}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1297}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, b) &= \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1298}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, b) &= -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1299}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, b) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z1300}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) &= -\frac{3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1301}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) &= \frac{3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}
\end{aligned}$$

$$\boxed{\text{z1302}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1303}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1304}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1305}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

• **A;A_002_2** : 'A'-'A' bond-cluster

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: **3b03c**

$$\boxed{\text{z54}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z55}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z56}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z57}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g) = \mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z58}} \quad \mathbb{G}_3^{(c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z59}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z119}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u) = -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z120}} \quad \mathbb{Q}_3^{(1,0;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z121}} \quad \mathbb{G}_0^{(c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z122}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z123}} \quad \mathbb{G}_4^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z124}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z241}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z242}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z243}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z244}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z245}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z246}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z247}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z248}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z249}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z250}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z251}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z252}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z371}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z372}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z373}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14}\left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{28}$$

$$\boxed{\text{z374}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14}\left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{84}$$

$$\boxed{\text{z375}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{3}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z376}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z377}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{14}\left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{84}$$

$$\boxed{\text{z378}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{14}\left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{28}$$

$$\boxed{\text{z379}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z380}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u) = \frac{\sqrt{3}\left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z381}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z382}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z799}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z800}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z801}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z802}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z803}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z804}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z805}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 1) = -\frac{3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z806}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z807}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z808}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z809}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\begin{aligned}
\boxed{\text{z810}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z811}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z812}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) &= \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z813}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) &= -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z814}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1) &= -\frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z815}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1) &= \frac{3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z816}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z817}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g) &= -\frac{3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z818}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g) &= \frac{3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z819}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z820}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z821}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z822}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, a) &= \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z823}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, b) &= \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}
\end{aligned}$$

$$\boxed{\text{z824}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z825}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1306}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z1307}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z1308}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z1309}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z1310}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z1311}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)\right)}{6}$$

$$\boxed{\text{z1312}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1) = \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1313}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1) = \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1314}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1) = \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1315}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1316}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$$\begin{aligned}
\boxed{\text{z1317}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} \\
\boxed{\text{z1318}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z1319}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z1320}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z1321}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z1322}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1) &= -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\boxed{\text{z1323}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{15} \\
\boxed{\text{z1324}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) &= \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} \\
\boxed{\text{z1325}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} \\
\boxed{\text{z1326}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} \\
\boxed{\text{z1327}} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z1328}} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\boxed{\text{z1329}} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z1330}} \quad \mathbb{G}_{2,1}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1331}} \quad \mathbb{G}_{2,2}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1332}} \quad \mathbb{G}_{2,3}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1333}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{21}$$

$$\boxed{\text{z1334}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{21}$$

$$\boxed{\text{z1335}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, a) = \frac{\sqrt{7} \left(-\sqrt{3} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5} \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{21}$$

$$\boxed{\text{z1336}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1337}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1338}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, b) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1339}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 1) = \frac{\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1340}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 1) = \frac{-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - 3 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1341}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 1) = \frac{\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - 3 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - 3 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1342}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{84}$$

$$\boxed{\text{z1343}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) + 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - 9 \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{84}$$

$$\boxed{\text{z1344}} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{7} \left(-\sqrt{15} \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - 9 \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15} \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + 9 \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + 12 \mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84}$$

$$\boxed{\text{z1345}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u) = \frac{\sqrt{2} \left(\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1346}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1347}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u) = \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) - 2 \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1348}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1349}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1350}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

• **A;A_002_3** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$

* wyckoff: **3c@3c**

$$\boxed{\text{z60}} \quad \mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z125}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z253}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z254}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z383}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z384}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z1351}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1352}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1353}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1354}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1355}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1356}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

• **A;A_002_3** : 'A'-'A' bond-cluster

* bra: $\langle s, \uparrow |, \langle s, \downarrow |$

* ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

* wyckoff: **3c@3c**

$$\boxed{\text{z61}} \quad \mathbb{Q}_0^{(c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z62}} \quad \mathbb{Q}_0^{(1,0;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z63}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{3} \left(\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z126}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z127}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{2} \left(\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z128}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z255}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{1,3}^{(a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z256}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z257}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z258}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z259}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(E_g) = \frac{\sqrt{3} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{1,3}^{(1,0;a)}(T_u) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z260}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z385}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z386}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2} \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z387}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z388}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z389}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2} \mathbb{G}_0^{(1,1;a)}(A_u) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z830}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z826}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z827}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z828}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z829}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z830}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z831}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g) = \frac{\sqrt{2} \left(-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z832}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z833}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z834}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z835}} \quad \mathbb{G}_{1,1}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z836}} \quad \mathbb{G}_{1,2}^{(c)}(T_g) = \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z837}} \quad \mathbb{G}_{1,3}^{(c)}(T_g) = \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\begin{aligned}
\text{z838} \quad \mathbb{G}_{1,1}^{(1,-1;c)}(T_g) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\text{z839} \quad \mathbb{G}_{1,2}^{(1,-1;c)}(T_g) &= \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z840} \quad \mathbb{G}_{1,3}^{(1,-1;c)}(T_g) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z841} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{5} \left(-3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\text{z842} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 1) &= -\frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z843} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 1) &= \frac{\sqrt{5} \left(3\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{15} \\
\text{z844} \quad \mathbb{G}_{3,1}^{(1,-1;c)}(T_g, 2) &= \frac{-\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} \\
\text{z845} \quad \mathbb{G}_{3,2}^{(1,-1;c)}(T_g, 2) &= \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} \\
\text{z846} \quad \mathbb{G}_{3,3}^{(1,-1;c)}(T_g, 2) &= \frac{-\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} \\
\text{z847} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\text{z848} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g) &= \frac{\sqrt{6} \left(-\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z849} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g) &= \frac{\sqrt{6} \left(\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z850} \quad \mathbb{G}_{1,1}^{(1,1;c)}(T_g) &= \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}
\end{aligned}$$

$$\boxed{\text{z851}} \quad \mathbb{G}_{1,2}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z852}} \quad \mathbb{G}_{1,3}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$$\boxed{\text{z1357}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1358}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1359}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z1360}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u, b) = \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1361}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1362}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1363}} \quad \mathbb{Q}_{3,1}^{(c)}(T_u, 2) = -\frac{3\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1364}} \quad \mathbb{Q}_{3,2}^{(c)}(T_u, 2) = \frac{3\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1365}} \quad \mathbb{Q}_{3,3}^{(c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1366}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1367}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = \frac{-3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z1368}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\begin{aligned}
\boxed{\text{z1369}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) &= \frac{-\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1370}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) &= -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1371}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z1372}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1373}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1374}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, a) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1375}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u, b) &= \frac{-\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1376}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u, b) &= -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1377}} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u, b) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3} \\
\boxed{\text{z1378}} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) &= -\frac{3\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1379}} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) &= \frac{3\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} \\
\boxed{\text{z1380}} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3} \\
\boxed{\text{z1381}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u) &= \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3} \\
\boxed{\text{z1382}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u) &= \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}
\end{aligned}$$

$$\boxed{\text{z1383}} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

• **A;A_002_3** : 'A'-A' bond-cluster

* bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$

* ket: $|p_x, \uparrow \rangle, |p_x, \downarrow \rangle, |p_y, \uparrow \rangle, |p_y, \downarrow \rangle, |p_z, \uparrow \rangle, |p_z, \downarrow \rangle$

* wyckoff: 3c@3c

$$\boxed{\text{z64}} \quad \mathbb{Q}_0^{(c)}(A_g, a) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z65}} \quad \mathbb{Q}_0^{(c)}(A_g, b) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z66}} \quad \mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z67}} \quad \mathbb{Q}_0^{(1,1;c)}(A_g) = \mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$$\boxed{\text{z68}} \quad \mathbb{G}_3^{(c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z69}} \quad \mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{2} \left(\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g) \right)}{2}$$

$$\boxed{\text{z129}} \quad \mathbb{Q}_3^{(1,-1;c)}(A_u) = -\frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z130}} \quad \mathbb{Q}_3^{(1,0;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z131}} \quad \mathbb{G}_0^{(c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z132}} \quad \mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z133}} \quad \mathbb{G}_4^{(1,-1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z134}} \quad \mathbb{G}_0^{(1,1;c)}(A_u) = \frac{\sqrt{3} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{3}$$

$$\boxed{\text{z261}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z262}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_0^{(a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z263}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z264}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z265}} \quad \mathbb{Q}_{2,1}^{(c)}(E_g, c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z266}} \quad \mathbb{Q}_{2,2}^{(c)}(E_g, c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z267}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z268}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2} \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_0^{(b)}(A_g)}{2}$$

$$\boxed{\text{z269}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g) - \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z270}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,2}^{(b)}(E_g) + \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z271}} \quad \mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z272}} \quad \mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2} \mathbb{Q}_0^{(1,1;a)}(A_g) \mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$$\boxed{\text{z391}} \quad \mathbb{G}_{2,1}^{(c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z392}} \quad \mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z393}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{28}$$

$$\boxed{\text{z394}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{14} \left(9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84}$$

$$\boxed{\text{z395}} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z396}} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z397}} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{14} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84}$$

$$\boxed{\text{z398}} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{14} \left(\sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) - 2\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{28}$$

$$\boxed{\text{z399}} \quad \mathbb{G}_{2,1}^{(1,0;c)}(E_u) = \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z400}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z401}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{3} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z402}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$$\boxed{\text{z853}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g) \mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z854}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z855}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z856}} \quad \mathbb{Q}_{2,1}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z857}} \quad \mathbb{Q}_{2,2}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z858}} \quad \mathbb{Q}_{2,3}^{(c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z859}} \quad \mathbb{Q}_{4,1}^{(c)}(T_g, 1) = -\frac{3\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z860}} \quad \mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{3\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z861}} \quad \mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z862}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z863}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z864}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z865}} \quad \mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - 3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z866}} \quad \mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z867}} \quad \mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z868}} \quad \mathbb{Q}_{4,1}^{(1,-1;c)}(T_g, 1) = -\frac{3\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z869}} \quad \mathbb{Q}_{4,2}^{(1,-1;c)}(T_g, 1) = \frac{3\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z870}} \quad \mathbb{Q}_{4,3}^{(1,-1;c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z871}} \quad \mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = -\frac{3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + \sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z872}} \quad \mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) - \sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z873}} \quad \mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z874}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z875}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z876}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_0^{(b)}(A_g)}{3}$$

$$\boxed{\text{z877}} \quad \mathbb{G}_{1,1}^{(1,0;c)}(T_g, b) = \frac{-\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z878}} \quad \mathbb{G}_{1,2}^{(1,0;c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g) + 3\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$$\boxed{\text{z879}} \quad \mathbb{G}_{1,3}^{(1,0;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$$

$$\boxed{\text{z1384}} \quad \mathbb{Q}_{1,1}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1385}} \quad \mathbb{Q}_{1,2}^{(c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1386}} \quad \mathbb{Q}_{1,3}^{(c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1387}} \quad \mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1388}} \quad \mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1389}} \quad \mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1390}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 1) = \frac{-3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1391}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 1) = \frac{3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1392}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 1) = \frac{-3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1393}} \quad \mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1394}} \quad \mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1395}} \quad \mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) \mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) \mathbb{T}_{1,1}^{(b)}(T_u) + 4\mathbb{M}_3^{(1,-1;a)}(A_g) \mathbb{T}_{1,3}^{(b)}(T_u)}{12}$$

$$\boxed{\text{z1396}} \quad \mathbb{Q}_{1,1}^{(1,0;c)}(T_u) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,1}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30}$$

$$\boxed{\text{z1397}} \quad \mathbb{Q}_{1,2}^{(1,0;c)}(T_u) = \frac{\sqrt{10} \left(-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30}$$

$$\begin{aligned}
\text{z1398} \quad \mathbb{Q}_{1,3}^{(1,0;c)}(T_u) &= \frac{\sqrt{10} \left(2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 3\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z1399} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 1) &= \frac{\sqrt{5} \left(-3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{30} \\
\text{z1400} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 1) &= -\frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 3\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{30} \\
\text{z1401} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 1) &= \frac{\sqrt{5} \left(3\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{15} \\
\text{z1402} \quad \mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) &= \frac{-\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - 2\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} \\
\text{z1403} \quad \mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) &= \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) + 2\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u) - 2\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} \\
\text{z1404} \quad \mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) &= \frac{-\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} \\
\text{z1405} \quad \mathbb{Q}_{1,1}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) - \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\text{z1406} \quad \mathbb{Q}_{1,2}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(-\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z1407} \quad \mathbb{Q}_{1,3}^{(1,1;c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z1408} \quad \mathbb{G}_{2,1}^{(c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\text{z1409} \quad \mathbb{G}_{2,2}^{(c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z1410} \quad \mathbb{G}_{2,3}^{(c)}(T_u) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}
\end{aligned}$$

$$\begin{aligned}
\text{z1411} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{21} \\
\text{z1412} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{21} \\
\text{z1413} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, a) &= \frac{\sqrt{7} \left(-\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - \sqrt{5}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + \sqrt{5}\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{21} \\
\text{z1414} \quad \mathbb{G}_{2,1}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6} \\
\text{z1415} \quad \mathbb{G}_{2,2}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z1416} \quad \mathbb{G}_{2,3}^{(1,-1;c)}(T_u, b) &= \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6} \\
\text{z1417} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 1) &= \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} \\
\text{z1418} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 1) &= \frac{-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) + \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - 3\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} \\
\text{z1419} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 1) &= \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 3\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - 3\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} \\
\text{z1420} \quad \mathbb{G}_{4,1}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) - 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) + 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u) \right)}{84} \\
\text{z1421} \quad \mathbb{G}_{4,2}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,3}^{(b)}(T_u) + 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,3}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) - 9\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{84} \\
\text{z1422} \quad \mathbb{G}_{4,3}^{(1,-1;c)}(T_u, 2) &= \frac{\sqrt{7} \left(-\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,2}^{(b)}(T_u) - 9\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,2}^{(b)}(T_u) - \sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{T}_{1,1}^{(b)}(T_u) + 9\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{T}_{1,1}^{(b)}(T_u) + 12\mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u) \right)}{84} \\
\text{z1423} \quad \mathbb{G}_{2,1}^{(1,0;c)}(T_u) &= \frac{\sqrt{2} \left(\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u) - \mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}
\end{aligned}$$

$$\boxed{\text{z1424}} \quad \mathbb{G}_{2,2}^{(1,0;c)}(T_u) = \frac{\sqrt{2} \left(-\sqrt{3} \mathbb{T}_{2,1}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,2}^{(b)}(T_u) - \mathbb{T}_{2,3}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1425}} \quad \mathbb{G}_{2,3}^{(1,0;c)}(T_u) = \frac{\sqrt{2} \left(-\mathbb{T}_{2,1}^{(1,0;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) - 2 \mathbb{T}_{2,2}^{(1,0;a)}(E_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{T}_{2,2}^{(1,0;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1426}} \quad \mathbb{G}_{2,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1427}} \quad \mathbb{G}_{2,2}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,3}^{(b)}(T_u) + \mathbb{M}_{1,3}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

$$\boxed{\text{z1428}} \quad \mathbb{G}_{2,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6} \left(\mathbb{M}_{1,1}^{(1,1;a)}(T_g) \mathbb{T}_{1,2}^{(b)}(T_u) + \mathbb{M}_{1,2}^{(1,1;a)}(T_g) \mathbb{T}_{1,1}^{(b)}(T_u) \right)}{6}$$

Atomic SAMB

- bra: $\langle s, \uparrow |, \langle s, \downarrow |$
- ket: $|s, \uparrow\rangle, |s, \downarrow\rangle$

$$\boxed{\text{x1}} \quad \mathbb{Q}_0^{(a)}(A_g) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$\boxed{\text{x2}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x3}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$$

$$\boxed{\text{x4}} \quad \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

- bra: $\langle s, \uparrow |, \langle s, \downarrow |$

- ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

$$\boxed{\text{x5}} \quad \mathbb{Q}_{1,1}^{(a)}(T_u) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x6}} \quad \mathbb{Q}_{1,2}^{(a)}(T_u) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x7}} \quad \mathbb{Q}_{1,3}^{(a)}(T_u) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$$\boxed{\text{x8}} \quad \mathbb{Q}_{1,1}^{(1,0;a)}(T_u) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x9}} \quad \mathbb{Q}_{1,2}^{(1,0;a)}(T_u) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x10}} \quad \mathbb{Q}_{1,3}^{(1,0;a)}(T_u) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x11}} \quad \mathbb{G}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$$

$$\boxed{\text{x12}} \quad \mathbb{G}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x13}} \quad \mathbb{G}_{2,1}^{(1,-1;a)}(T_u) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x14}} \quad \mathbb{G}_{2,2}^{(1,-1;a)}(T_u) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x15}} \quad \mathbb{G}_{2,3}^{(1,-1;a)}(T_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x16}} \quad \mathbb{G}_0^{(1,1;a)}(A_u) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$$

$$\boxed{\text{x17}} \quad \mathbb{M}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{6} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x18}} \quad \mathbb{M}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x19}} \quad \mathbb{M}_{2,1}^{(1,-1;a)}(T_u) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x20}} \quad \mathbb{M}_{2,2}^{(1,-1;a)}(T_u) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x21}} \quad \mathbb{M}_{2,3}^{(1,-1;a)}(T_u) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x22}} \quad \mathbb{M}_0^{(1,1;a)}(A_u) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$$

$$\boxed{\text{x23}} \quad \mathbb{T}_{1,1}^{(a)}(T_u) = \begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x24}} \quad \mathbb{T}_{1,2}^{(a)}(T_u) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x25}} \quad \mathbb{T}_{1,3}^{(a)}(T_u) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$$

$$\boxed{\text{x26}} \quad \mathbb{T}_{1,1}^{(1,0;a)}(T_u) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x27}} \quad \mathbb{T}_{1,2}^{(1,0;a)}(T_u) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$$

$$\boxed{\text{x28}} \quad \mathbb{T}_{1,3}^{(1,0;a)}(T_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

- bra: $\langle p_x, \uparrow |, \langle p_x, \downarrow |, \langle p_y, \uparrow |, \langle p_y, \downarrow |, \langle p_z, \uparrow |, \langle p_z, \downarrow |$
- ket: $|p_x, \uparrow\rangle, |p_x, \downarrow\rangle, |p_y, \uparrow\rangle, |p_y, \downarrow\rangle, |p_z, \uparrow\rangle, |p_z, \downarrow\rangle$

$$\boxed{\text{x29}} \quad \mathbb{Q}_0^{(a)}(A_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x30}} \quad \mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$$

$$\boxed{\text{x31}} \quad \mathbb{Q}_{2,2}^{(a)}(E_g) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x32}} \quad \mathbb{Q}_{2,1}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x33}} \quad \mathbb{Q}_{2,2}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x34}} \quad \mathbb{Q}_{2,3}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x35}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x36}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x37}} \quad \mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x38}} \quad \mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x39}} \quad \mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x40}} \quad \mathbb{Q}_0^{(1,1;a)}(A_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x41}} \quad \mathbb{G}_{1,1}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x42}} \quad \mathbb{G}_{1,2}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x43}} \quad \mathbb{G}_{1,3}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x44}} \quad \mathbb{M}_{1,1}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x45}} \quad \mathbb{M}_{1,2}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x46}} \quad \mathbb{M}_{1,3}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x47}} \quad \mathbb{M}_3^{(1,-1;a)}(A_g) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x48}} \quad \mathbb{M}_{1,1}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x49}} \quad \mathbb{M}_{1,2}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x50}} \quad \mathbb{M}_{1,3}^{(1,-1;a)}(T_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$$

$$\boxed{\text{x51}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{5}}{5} & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{5}}{5} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x52}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1) = \begin{bmatrix} 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{5} & -\frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$$

$$\boxed{\text{x53}} \quad \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$$

$$\boxed{\text{x54}} \quad \mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x55}} \quad \mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x56}} \quad \mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2) = \begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x57}} \quad \mathbb{M}_{1,1}^{(1,1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$$

$$\boxed{\text{x58}} \quad \mathbb{M}_{1,2}^{(1,1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$$

$$\boxed{\text{x59}} \quad \mathbb{M}_{1,3}^{(1,1;a)}(T_g) = \begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$$

$$\boxed{\text{x60}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x61}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{x62}} \quad \mathbb{T}_{2,1}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x63}} \quad \mathbb{T}_{2,2}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$$

$$\boxed{\text{x64}} \quad \mathbb{T}_{2,3}^{(1,0;a)}(T_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$$

Cluster SAMB

- Site cluster

** Wyckoff: 3d

$$\boxed{\text{y1}} \quad \mathbb{Q}_0^{(s)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y2}} \quad \mathbb{Q}_{2,1}^{(s)}(E_g) = \left[-\frac{10\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{2778}, \frac{43\sqrt{2778}}{2778} \right]$$

$$\boxed{\text{y3}} \quad \mathbb{Q}_{2,2}^{(s)}(E_g) = \left[\frac{11\sqrt{926}}{463}, -\frac{21\sqrt{926}}{926}, -\frac{\sqrt{926}}{926} \right]$$

- Bond cluster

** Wyckoff: 3a@1a

$$\boxed{\text{y4}} \quad \mathbb{Q}_0^{(b)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y5}} \quad \mathbb{Q}_{2,1}^{(b)}(E_g) = \left[-\frac{10\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{2778}, \frac{43\sqrt{2778}}{2778} \right]$$

$$\boxed{\text{y6}} \quad \mathbb{Q}_{2,2}^{(b)}(E_g) = \left[\frac{11\sqrt{926}}{463}, -\frac{21\sqrt{926}}{926}, -\frac{\sqrt{926}}{926} \right]$$

$$\boxed{\text{y7}} \quad \mathbb{T}_{1,1}^{(b)}(T_u) = [i, 0, 0]$$

$$\boxed{\text{y8}} \quad \mathbb{T}_{1,2}^{(b)}(T_u) = [0, i, 0]$$

$$\boxed{\text{y9}} \quad \mathbb{T}_{1,3}^{(b)}(T_u) = [0, 0, i]$$

** Wyckoff: 3b@3c

$$\boxed{\text{y10}} \quad \mathbb{Q}_0^{(b)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y11}} \quad \mathbb{Q}_{2,1}^{(b)}(E_g) = \left[-\frac{10\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{2778}, \frac{43\sqrt{2778}}{2778} \right]$$

$$\boxed{\text{y12}} \quad \mathbb{Q}_{2,2}^{(b)}(E_g) = \left[\frac{11\sqrt{926}}{463}, -\frac{21\sqrt{926}}{926}, -\frac{\sqrt{926}}{926} \right]$$

$$\boxed{\text{y13}} \quad \mathbb{T}_{1,1}^{(b)}(T_u) = [0, i, 0]$$

$$\boxed{\text{y14}} \quad \mathbb{T}_{1,2}^{(b)}(T_u) = [0, 0, i]$$

$$\boxed{\text{y15}} \quad \mathbb{T}_{1,3}^{(b)}(T_u) = [i, 0, 0]$$

** Wyckoff: 3c@3c

$$\boxed{\text{y16}} \quad \mathbb{Q}_0^{(b)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right]$$

$$\boxed{\text{y17}} \quad \mathbb{Q}_{2,1}^{(b)}(E_g) = \left[-\frac{10\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{2778}, \frac{43\sqrt{2778}}{2778} \right]$$

$$\boxed{\text{y18}} \quad \mathbb{Q}_{2,2}^{(b)}(E_g) = \left[\frac{11\sqrt{926}}{463}, -\frac{21\sqrt{926}}{926}, -\frac{\sqrt{926}}{926} \right]$$

$$\boxed{\text{y19}} \quad \mathbb{T}_{1,1}^{(b)}(T_u) = [0, 0, i]$$

$$\boxed{\text{y20}} \quad \mathbb{T}_{1,2}^{(b)}(T_u) = [i, 0, 0]$$

$$\boxed{\text{y21}} \quad \mathbb{T}_{1,3}^{(b)}(T_u) = [0, i, 0]$$

** Wyckoff: 12a012j

$$\boxed{\text{y22}} \quad \mathbb{Q}_0^{(b)}(A_g) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6} \right]$$

$$\boxed{\text{y23}} \quad \mathbb{T}_0^{(b)}(A_g) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6} \right]$$

$$\boxed{\text{y24}} \quad \mathbb{Q}_{2,1}^{(b)}(E_g) = \left[-\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556} \right]$$

$$\boxed{\text{y25}} \quad \mathbb{Q}_{2,2}^{(b)}(E_g) = \left[\frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852} \right]$$

$$\boxed{\text{y26}} \quad \mathbb{T}_{2,1}^{(b)}(E_g) = \left[-\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556} \right]$$

$$\boxed{\text{y27}} \quad \mathbb{T}_{2,2}^{(b)}(E_g) = \left[\frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852} \right]$$

$$\boxed{\text{y28}} \quad \mathbb{M}_{1,1}^{(b)}(T_g) = \left[\frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, 0, 0, 0, 0, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y29}} \quad \mathbb{M}_{1,2}^{(b)}(T_g) = \left[0, 0, 0, 0, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y30}} \quad \mathbb{M}_{1,3}^{(b)}(T_g) = \left[0, 0, 0, 0, 0, 0, 0, 0, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2} \right]$$

$$\boxed{\text{y31}} \quad \mathbb{Q}_{2,1}^{(b)}(T_g) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y32}} \quad \mathbb{Q}_{2,2}^{(b)}(T_g) = \left[0, 0, 0, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y33}} \quad \mathbb{Q}_{2,3}^{(b)}(T_g) = \left[0, 0, 0, 0, 0, 0, 0, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right]$$

$$\boxed{\text{y34}} \quad \mathbb{Q}_{1,1}^{(b)}(T_u) = \left[0, 0, 0, 0, \frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14} \right]$$

$$\boxed{\text{y35}} \quad \mathbb{Q}_{1,2}^{(b)}(T_u) = \left[\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, 0, 0, 0, 0, \frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7} \right]$$

$$\boxed{\text{y36}} \quad \mathbb{Q}_{1,3}^{(b)}(T_u) = \left[\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y37}} \quad \mathbb{T}_{1,1}^{(b)}(T_u, a) = \left[0, 0, 0, 0, \frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14} \right]$$

$$\boxed{\text{y38}} \quad \mathbb{T}_{1,2}^{(b)}(T_u, a) = \left[\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, 0, 0, 0, 0, \frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7} \right]$$

$$\boxed{\text{y39}} \quad \mathbb{T}_{1,3}^{(b)}(T_u, a) = \left[\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y40}} \quad \mathbb{T}_{1,1}^{(b)}(T_u, b) = \left[0, 0, 0, 0, \frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7} \right]$$

$$\boxed{\text{y41}} \quad \mathbb{T}_{1,2}^{(b)}(T_u, b) = \left[-\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, 0, 0, 0, 0, \frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14} \right]$$

$$\boxed{\text{y42}} \quad \mathbb{T}_{1,3}^{(b)}(T_u, b) = \left[\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, 0, 0, 0, 0 \right]$$

$$\boxed{\text{y43}} \quad \mathbb{Q}_{3,1}^{(b)}(T_u, 1) = \left[0, 0, 0, 0, \frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7} \right]$$

$$\boxed{\text{y44}} \quad \mathbb{Q}_{3,2}^{(b)}(T_u, 1) = \left[-\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, 0, 0, 0, 0, \frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14} \right]$$

$$\boxed{\text{y45}} \quad \mathbb{Q}_{3,3}^{(b)}(T_u, 1) = \left[\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, 0, 0, 0, 0 \right]$$

Table 5: Orbital of each site

#	site	orbital
1	A	$ s, \uparrow\rangle, s, \downarrow\rangle, p_x, \uparrow\rangle, p_x, \downarrow\rangle, p_y, \uparrow\rangle, p_y, \downarrow\rangle, p_z, \uparrow\rangle, p_z, \downarrow\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	A	A	[1,2]	[s,p]	[s,p]

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'A' (#1) site cluster (3d), mmm..

SL	position (s)	mapping
1	[0.50000, 0.00000, 0.00000]	[1,2,3,4,13,14,15,16]
2	[0.00000, 0.50000, 0.00000]	[5,6,7,8,17,18,19,20]
3	[0.00000, 0.00000, 0.50000]	[9,10,11,12,21,22,23,24]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'A'-'A' [1] (#1) bond cluster (12a@12j), D, $|\mathbf{v}| = 0.70711$ (cartesian)

SL	vector ($\mathbf{v}$)	center ($\mathbf{c}$)	mapping	head	tail	$\mathbf{R}$ (primitive)
1	[0.00000, -0.50000, -0.50000]	[0.00000, 0.25000, 0.75000]	[1,16]	(3,1)	(2,1)	[0,0,1]
2	[0.00000, 0.50000, -0.50000]	[0.00000, 0.75000, 0.75000]	[2,15]	(3,1)	(2,1)	[0,-1,1]
3	[0.00000, -0.50000, 0.50000]	[0.00000, 0.25000, 0.25000]	[3,14]	(3,1)	(2,1)	[0,0,0]
4	[0.00000, 0.50000, 0.50000]	[0.00000, 0.75000, 0.25000]	[4,13]	(3,1)	(2,1)	[0,-1,0]
5	[-0.50000, 0.00000, -0.50000]	[0.75000, 0.00000, 0.25000]	[5,20]	(1,1)	(3,1)	[1,0,0]
6	[-0.50000, 0.00000, 0.50000]	[0.75000, 0.00000, 0.75000]	[6,19]	(1,1)	(3,1)	[1,0,-1]
7	[0.50000, 0.00000, -0.50000]	[0.25000, 0.00000, 0.25000]	[7,18]	(1,1)	(3,1)	[0,0,0]
8	[0.50000, 0.00000, 0.50000]	[0.25000, 0.00000, 0.75000]	[8,17]	(1,1)	(3,1)	[0,0,-1]
9	[-0.50000, -0.50000, 0.00000]	[0.25000, 0.75000, 0.00000]	[9,24]	(2,1)	(1,1)	[0,1,0]
10	[0.50000, -0.50000, 0.00000]	[0.75000, 0.75000, 0.00000]	[10,23]	(2,1)	(1,1)	[-1,1,0]
11	[-0.50000, 0.50000, 0.00000]	[0.25000, 0.25000, 0.00000]	[11,22]	(2,1)	(1,1)	[0,0,0]
12	[0.50000, 0.50000, 0.00000]	[0.75000, 0.25000, 0.00000]	[12,21]	(2,1)	(1,1)	[-1,0,0]

Table 9: 2-th 'A'-'A' [1] (#2) bond cluster (3a@1a), ND, $|\mathbf{v}| = 1.0$ (cartesian)

SL	vector ($\mathbf{v}$)	center ($\mathbf{c}$)	mapping	head	tail	$\mathbf{R}$ (primitive)
1	[-1.00000, 0.00000, 0.00000]	[0.00000, 0.00000, 0.00000]	[1,-2,-3,4,-13,14,15,-16]	(1,1)	(1,1)	[1,0,0]

continued ...

Table 9

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
2	[0.00000, -1.00000, 0.00000]	[0.00000, 0.00000, 0.00000]	[5, -6, -7, 8, -17, 18, 19, -20]	(2, 1)	(2, 1)	[0, 1, 0]
3	[0.00000, 0.00000, -1.00000]	[0.00000, 0.00000, 0.00000]	[9, -10, -11, 12, -21, 22, 23, -24]	(3, 1)	(3, 1)	[0, 0, 1]

Table 10: 2-th 'A'-'A' [2] (#3) bond cluster (3b@3c), ND, $|\boldsymbol{v}|=1.0$ (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000, 0.00000, -1.00000]	[0.00000, 0.50000, 0.50000]	[1, 2, -3, -4, -13, -14, 15, 16]	(2, 1)	(2, 1)	[0, 0, 1]
2	[-1.00000, 0.00000, 0.00000]	[0.50000, 0.00000, 0.50000]	[5, 6, -7, -8, -17, -18, 19, 20]	(3, 1)	(3, 1)	[1, 0, 0]
3	[0.00000, -1.00000, 0.00000]	[0.50000, 0.50000, 0.00000]	[9, 10, -11, -12, -21, -22, 23, 24]	(1, 1)	(1, 1)	[0, 1, 0]

Table 11: 2-th 'A'-'A' [3] (#4) bond cluster (3c@3c), ND, $|\boldsymbol{v}|=1.0$ (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000, -1.00000, 0.00000]	[0.00000, 0.50000, 0.50000]	[1, -2, 3, -4, -13, 14, -15, 16]	(3, 1)	(3, 1)	[0, 1, 0]
2	[0.00000, 0.00000, -1.00000]	[0.50000, 0.00000, 0.50000]	[5, -6, 7, -8, -17, 18, -19, 20]	(1, 1)	(1, 1)	[0, 0, 1]
3	[-1.00000, 0.00000, 0.00000]	[0.50000, 0.50000, 0.00000]	[9, -10, 11, -12, -21, 22, -23, 24]	(2, 1)	(2, 1)	[1, 0, 0]